Aerospace Propulsion
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General Electric GE9X

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1 Introduction

The General Electric GE9X isn't just an innovation icon, as one of the most advanced engines ever made: it's a testament to the aviation sector's unwavering pursuit of excellence. Its development symbolizes a culmination of aeronautical engine technology progress, showcasing human ingenuity and engineering prowess.

Born from a vision to redefine efficiency and performance standards in commercial aviation, the GE9X is a high-bypass turbofan tailored exclusively for the Boeing 777X[1] by GE Aerospace in years of research, development, and testing. The GE9X journey began in February 2012 when General Electric embarked on enhancing its renowned GE90 engine to power the next-generation -8/-9 variants of the Boeing 777X and has been marked by significant milestones, from the inaugural ground test in April 2016, to when the GE9X shattered world records, showcasing unprecedented thrust levels and underscoring its reliability and performance in November 2017. March 13, 2018 marked the historic maiden flight, and subsequently it received Federal Aviation Administration (FAA) type certification on September 25, 2020. It is such rigorous testing and refinement that have positioned it as a pinnacle of engineering excellence, set to redefine commercial air travel's future.

The debut of the GE9X on the Boeing 777X signals a new era where cutting-edge technology meets efficiency and performance. As the aviation industry evolves, the GE9X inspires progress, shaping the future of air transportation for generations.

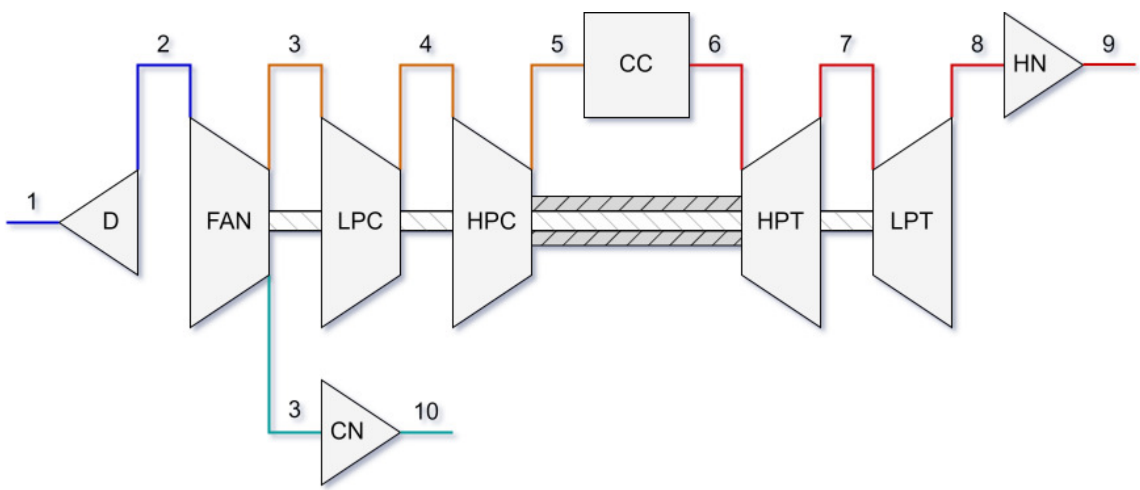

Figure 1: Engine scheme

[1]An insight into the aircraft can be found in the appendix

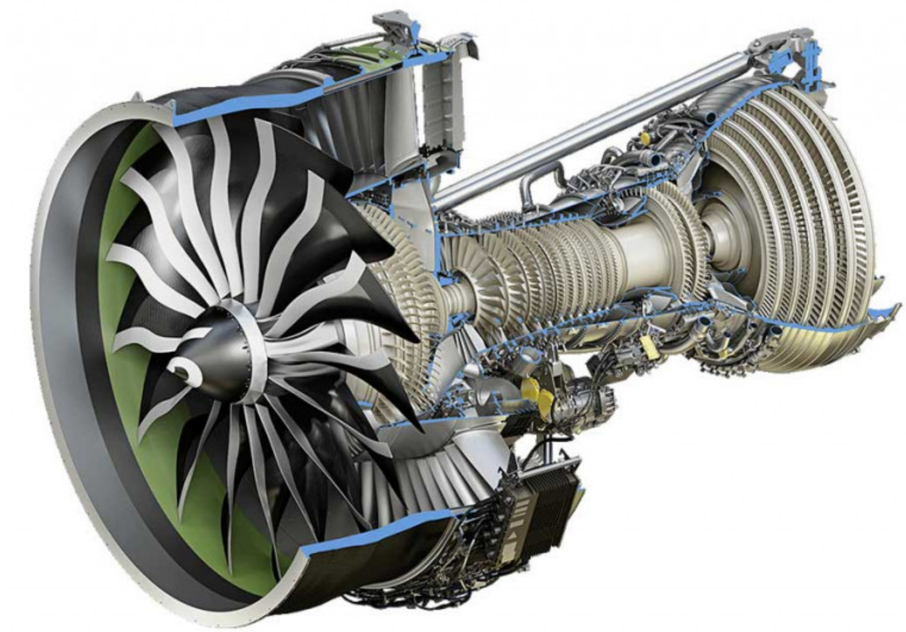

Figure 2: GE9X Engine detail

The purpose of this paper is to elucidate the fundamental operation of the engine, leveraging the knowledge acquired during our coursework. This analysis is conducted within the scope of our academic capabilities, aiming to provide a comprehensive yet accessible exploration of the GE9X's performance characteristics. To achieve this goal, an idealized model of the engine operating in leveled flight and cruise speed is employed, utilizing an ideal gas cycle corrected with appropriate efficiencies. The report traces the airflow through the various engine components and examines the innovative technologies implemented within them.

Finally, a critical analysis of the results is presented, based on the main performance parameters, in order to understand the qualities that position this engine at the pinnacle of propulsion systems within the field of commercial aviation.

2 Known data and Symbols

Symbol	Description	Value	Unit
P_{amb}	Standard pressure at sea level	101325	$[Pa]$
T_{amb}	Standard temperature at sea level	288.15	$[K]$
λ	Troposphere temperature lapse coefficient	-0.0065	$[K/m]$
$\bar{z}$	Tropopause altitude	11000	$[m]$
$\bar{P}$	Pressure at $\bar{z}$	22600	$[Pa]$
$\bar{T}$	Temperature at $\bar{z}$	216.5	$[K]$
z_{cruise}	Cruise altitude	39000	$[ft]$
M	Cruise speed Mach number	0.85	$[-]$
R	Gas constant	287	$\left[\frac{J}{kg \cdot K}\right]$
D_{fan}	Fan diameter	3.4	$[m]$
ω	Fan rotation speed	2510	$[rpm]$
BPR	Bypass ratio	9.9	$[-]$
β_{tot}	Total compression ratio (HPC-Amb)	60	$[-]$
β_c	Compressors compression ratio (HPC-LPC)	27	$[-]$
n_{lpc}	LPC stages	3	$[-]$
n_{hpc}	HPC stages	11	$[-]$
Q	Fuel heating value	42798	$\left[\frac{kJ}{K}\right]$
$T_{cc,max}$	Max temperature in Combustion Chamber	1588.71	$[K]$
$T_{T,max}$	Max temperature in turbines	1365.15	$[K]$
Symbol	**Efficiencies**	**Value**	
Π_d	Intake pressure efficiency	0.98	
η_f	Fan polytropic efficiency	0.93	
$\eta_{f,m}$	Fan mechanic efficiency	0.99	
η_{lpc}	LPC polytropic efficiency	0.91	
$\eta_{lpc,m}$	LPC mechanic efficiency	0.99	
η_{hpc}	HPC polytropic efficiency	0.91	
$\eta_{hpc,m}$	HPC mechanic efficiency	0.99	
η_{cc}	Combustion chamber polytropic efficiency	0.99	
Π_{cc}	Combustion chamber pressure efficiency	0.95	
η_{lpt}	LPT polytropic efficiency	0.93	
$\eta_{lpt,m}$	LPT mechanic efficiency	0.99	
η_{hpt}	HPT polytropic efficiency	0.93	
$\eta_{hpt,m}$	HPT mechanic efficiency	0.99	
η_{hn}	Hot nozzle isentropic efficiency	0.95	
η_{cn}	Cold nozzle isentropic efficiency	0.95	

Symbol	**Description**	**Unit**
P	Static pressure	$[Pa]$
T	Static temperature	$[K]$
P_0	Total pressure	$[Pa]$
T_0	Total temperature	$[K]$
$T_{0,s}$	Isentropic total temperature	$[K]$
a	Sound speed	$[m/s]$
Δh_0	Total specific enthalpy exchange	$[J/kg]$
h	specific enthalpy exchange	$[J/kg]$
β_d	Inlet compression ratio	$[-]$
β_f	Fan compression ratio	$[-]$
c	chord length	$[m]$
Re	Reynolds number	$[-]$
ν	viscosity	$[m^2/s]$
NPR	Nozzle Pressure Ratio	$[-]$
$F_{DeLaval}$	De Laval nozzle thrust	$[N]$
F_{conv}	Convergent nozzle thrust	$[N]$
PWR	Power	$[W]$
u	absolute velocity	$[m/s]$
W	relative velocity	$[m/s]$
α	Absolute velocity angle	$[^o]$
β	Relative velocity angle	$[^o]$
O/F	Air on fuel flow rate ratio	$[-]$

Table 1: Data sheet[1]

3 Inlet

Inlet (or diffuser) is a component that performs several crucial functions in the intake and compression process of the air. Its main function is to convert the kinetic energy of the air into pressure energy, preparing it for entry into the compressor. Important functions of the inlet are:

- *Slowing down and spreading the airflow*: at the engine inlet, the air is usually moving at high speed. The diffuser is designed to slow down this airflow and spread it out, thus increasing its static pressure. This process occurs through an increase in the cross-sectional area of the flow,according to the principal of conservation of mass $A_1V_1 = A_2V_2$, where A is the cross-sectional area and V is the velocity

- *Directing the flow*: the diffuser helps to direct the airflow in the correct direction, guiding it towards the compressor. This ensures an uniform and well directed flow, improving the overall efficiency of the engine.

- *Reducing turbulence*: the diffuser can also help to reduce turbulence in the incoming air, ensuring a more stable and uniform flow.

- *Increasing air pressure*: by converting kinetic energy into pressure energy, the diffuser increases the air pressure entering the compressor. This is essential to ensure that the compressor receives air at a sufficiently high pressure to start the compressor process.

3.1 Analytic steps for Inlet resolution

The calculations refer to an ideal model where air entering the inlet is considered to flow parallel to the engine axis and at the same speed as cruise speed, basically any spillage caused by the rotating fan is removed. Standard atmosphere model is used to calculated ambient conditions in below troposphere altitudes

$$T = T_{amb} + \lambda z \tag{3.1.1}$$

$$P = P_{amb} \left(\frac{T}{T_{amb}} \right)^{-\frac{g}{R\lambda}} \tag{3.1.2}$$

While above $\bar{z}$, tropopause, ambient pressure and temperature are calculated using

$$T = \bar{T} \tag{3.1.3}$$

$$P = \bar{P}^{\left(-g \frac{z_{cruise} - \bar{z}}{RT}\right)} \tag{3.1.4}$$

Where z_{cruise} is expressed in meters. Then the total temperature and pressure are found using Mach number

$$T_{0,in} = T \left(1 + \left(\frac{\gamma - 1}{2} \right) M^2 \right) \tag{3.1.5}$$

$$P_{0,in} = P \left(1 + \left(\frac{\gamma - 1}{2} \right) M^2 \right)^{\frac{\gamma}{\gamma - 1}} \tag{3.1.6}$$

The air density at inlet section can be calculated using perfect gas law

$$\rho = \frac{P}{RT} \tag{3.1.7}$$

Now it is immediate to find the air speed entering the engine using Mach number definition through sound speed a

$$u_{flight} = Ma = M\sqrt{\gamma RT} \tag{3.1.8}$$

From this point, assuming inlet section to be almost identical to fan section, the air flow rate is

$$\dot{m}_a = \rho u_{flight} \left(\frac{\pi}{4}\right) (D^2_{fan}) \tag{3.1.9}$$

Expansion in inlet is an adiabatic phenomenon

$$T_{0,out} = T_{0,in} \tag{3.1.10}$$

Thanks to the inlet pressure efficiency Π_d it's possible to calculate the total pressure at the exit of the intake

$$P_{0,out} = \Pi_d P_{0,in} \tag{3.1.11}$$

Finally the compression ratio of the diffuser is

$$\beta_d = \frac{P_{0,out}}{P} \tag{3.1.12}$$

4 Fan

Figure 3: GE9X Fan

A fan is a machine that moves air by generating a small pressure rise. A general rule is that the pressure ratio should not exceed 1.25 $\left(\frac{P_{02}}{P_{01}} < 1.25\right)$. When the pressure ratio exceeds 2 $\left(\frac{P_{02}}{P_{01}} > 2\right)$, the machine is referred to as a compressor. The fan of the GE9X engine has a diameter of 3.4 m . It consists of a single stage and has 16 blades, which is fewer than its predecessor, the GE90, which had 22 blades [1]. Additionally, the fan is connected to the low-pressure turbine, through a RPM reducer, meaning that part of the energy produced by the turbine is used to power the fan.

4.1 Specific heat vs Temperature

The specific heat of air varies appreciably with temperature. This effect is important in the study and sizing of turbomachines as it influences their fluid dynamics. For the analysis of the variability of specific heat, the effects related to pressure variation are neglected, and the pressure is assumed to be at standard conditions.[2] Using the polyfit function in Matlab, the best approximation is found to be a fourth-degree polynomial with the following regression coefficients:

α	β	γ	δ	ϵ
1.057429683	-0.000437146	1.0987E-06	-7.47122E-10	1.69E-13

The regression function is

$$c_p(T) = \alpha + \beta T + \gamma T^2 + \delta T^3 + \epsilon T^4 \tag{4.1.1}$$

the polynomial function's trend, with the tabulated values, is shown in the figure 4.

[2] 15°C, 1 atm

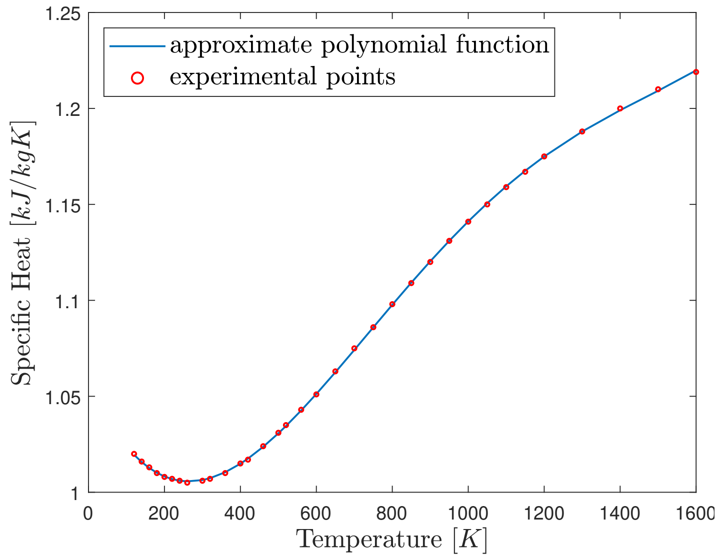

Figure 4: c_p vs T

This model for the air specific heat is implemented into the various stages regarding air coefficients, while the exhaust gases' c_p variations with the temperature are dismissable.

4.2 Analytic steps for Fan resolution

To calculate the fan compression ratio β_f, note that the product of the compression ratios of inlet, fan and compressors has to reach the total compression ratio β_{tot}

$$\beta_{tot} = \beta_f \beta_c \beta_d \rightarrow \beta_f = \frac{\beta_{tot}}{\beta_c \beta_d} \tag{4.2.1}$$

Where β_d is the same calculated in the inlet resolution section (3.1). Now, supposing polytropic compression, it's possible to calculate the total temperature of air leaving the fan

$$T_{0,out,s} = T_{0,in} \left(\beta_f\right)^{\frac{\gamma-1}{\gamma}} \tag{4.2.2}$$

It's possible to correct this value and obtain the real one by using the polytropic efficiency of the fan, η_f

$$T_{0,out} = T_{0,in} + \frac{\left(T_{0,out,s} - T_{0,in}\right)}{\eta_f} \tag{4.2.3}$$

The pressure at the exit of the fan is

$$P_{0,out} = \beta_f P_{0,in} \tag{4.2.4}$$

Finally power required to run the fan is

$$PWR = \frac{1}{\eta_{f,m}} c_p \dot{m}_a \left(T_{0,out} - T_{0,in}\right) \tag{4.2.5}$$

4.3 Sizing using Velocity triangles

The initial goal is to determine the Mach number at the fan inlet. This is possible starting from the conservation of mass

$$\dot{m} = \rho A u \rightarrow u = \frac{\dot{m}}{\rho A} = Ma \tag{4.3.1}$$

resulting in the following equation,

$$\frac{\dot{m}R}{A\frac{P_0}{\left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{\gamma}{\gamma-1}}}}\frac{T_0}{\left(1+\frac{\gamma-1}{2}M^2\right)} = M\sqrt{(\gamma-1)c_p\left(1+\frac{\gamma-1}{2}M^2\right)^{-1}T_0} \tag{4.3.2}$$

which is solvable through numerical methods (bisection, Newton, and fixed point).

The next step is the fan sizing. It start by assuming the hub to tip ratio value, which for fans could be 0.3. Then the values at the inlet are calculated

$$\begin{cases} T_1 = \frac{T_{01}}{1+\frac{\gamma-1}{2}M^2} \\ P_1 = P_{01}\left(\frac{T_1}{T_{01}}\right)^{\frac{\gamma}{\gamma-1}} \\ \rho_1 = \frac{P_1}{RT_1} \\ a_1 = \sqrt{\gamma R T_1} \\ u_1 = Ma_1 \end{cases} \tag{4.3.3}$$

the inlet area can be found through this relation

$$A_1 = \pi(r_{tip,1}^2 - r_{hub,1}^2) = \pi r_{tip,1}^2\left[1-\left(\frac{r_{hub,1}}{r_{tip,1}}\right)^2\right] \tag{4.3.4}$$

By using the conservation of mass equation, the tip radius can be calculated

$$r_{tip,1} = \sqrt{\frac{\dot{m}}{\pi\rho_1 u_1\left[1-\left(\frac{r_{hub,1}}{r_{tip,1}}\right)^2\right]}} \tag{4.3.5}$$

Consequently, the hub radius, inlet section, mean radius and blade height are obtained

$$r_m = \frac{r_{hub}+r_{tip}}{2} \rightarrow h_1 = r_{tip} - r_{hub} \tag{4.3.6}$$

The rotational speed is correlated with the tip speed and the tip radius as follows

$$U_{tip,1} = \omega r_{tip,1} \tag{4.3.7}$$

the relative Mach number at the inlet is verified

$$M_{1,rel} = \frac{w_1}{a_1} = \frac{\sqrt{u_1^2 + U_{tip}^2}}{\sqrt{\gamma R T_1}} \tag{4.3.8}$$

Now the values at the exit are calculated

$$\begin{cases} T_2 = T_{02} - \frac{u_1^2}{2c_p} \\ P_2 = P_{02} \left(\frac{T_2}{T_{02}}\right)^{\frac{\gamma}{\gamma-1}} \\ \rho_2 = \frac{P_2}{RT_2} \\ A_2 = \frac{\dot{m}}{\rho_2 u_1} \end{cases} \tag{4.3.9}$$

The total enthalpy change is

$$(\Delta h_0)_{tot} = c_p(T_{02} - T_{01}) \tag{4.3.10}$$

Now, a choice can be made among different operational strategies:

- Keeping the tip radius constant, which minimizes the blade dimensions in the later stages and requires high hub to tip ratios.
- Keeping the hub radius constant, in this case the blade dimensions become significantly larger and the average velocities decrease.
- Keeping the mean radius constant, which is a compromise solution, and therefore, the one that was chosen in this paper.

Following the choice that has been made, the blade height at the exit can be estimated

$$A_2 = \pi(r_{tip,2}^2 - r_{hub,2}^2) = \pi(r_{tip,2} + r_{hub,2})(r_{tip,2} - r_{hub,2}) = 2\pi r_m h_2 \tag{4.3.11}$$

hence h is calculated tip and hub radius can be easily obtained.

$$r_{tip,2} = r_m + \frac{h_2}{2} \ , \ r_{hub,2} = r_m - \frac{h_2}{2} \tag{4.3.12}$$

As specified in the introduction, the fan consists of only one stage. Therefore, the total enthalpy variation for this stage corresponds to the enthalpy variation of the entire fan calculated previously. Now, $\alpha_1 = 0$ (angle of absolute velocities) is assumed. This means that the inlet velocity is directed along the fan axis. The tangential velocity at the mean radius is calculated as follows:

$$U_m = \omega r_m \tag{4.3.13}$$

exit absolute velocity at the mean radius :

$$\begin{cases} u_{2m,\theta} = \frac{\Delta h_0}{U_m} \\ u_{2m} = \sqrt{u_1^2 + u_{2m,\theta}^2} \end{cases} \tag{4.3.14}$$

the exit relative velocity at the mean radius :

$$\begin{cases} W_{2m,\theta} = U_m - u_{2m,\theta} \\ W_{2m} = \sqrt{u_1^2 + W_{2m,\theta}^2} \end{cases} \tag{4.3.15}$$

the inlet relative velocity at the mean radius :

$$\begin{cases} W_{1m} = \sqrt{u_1^2 + U_m^2} \\ W_{1m,\theta} = U_m \end{cases} \tag{4.3.16}$$

It is necessary to ensure the flow deflection angle not to be excessive in order to avoid the detachment of the boundary layer from the back of the blade profiles. De Haller's empirical criterion can be used to verify the absence of separation

$$\frac{W_{2m}}{W_{1m}} \geq 0.72 \ , \ \frac{u_1}{U_{2m}} \geq 0.72 \tag{4.3.17}$$

and it is verified. Now, the blade angles can be evaluated

$$\begin{cases} \tan\beta_1 = \frac{W_{1m,\theta}}{u_1} \rightarrow \beta_1 \\ \tan\beta_2 = \frac{W_{2m,\theta}}{u_1} \rightarrow \beta_2 \\ \tan\alpha_2 = \frac{U_{2m,\theta}}{u_1} \rightarrow \alpha_2 \end{cases} \tag{4.3.18}$$

From here on, the fan reaction degree is

$$^{o}R = \left(\frac{u_1}{2U_m}\right)(\tan\beta_1 + \tan\beta_2) \tag{4.3.19}$$

if compared to turbomachinery, this value is higher in the fan. The loading coefficient and the flow coefficient, are found as follows

$$\begin{cases} \phi = \frac{u_1}{U_m} \\ \psi = 1 - \phi(\tan\beta_2 - \tan\alpha_1) \end{cases} \tag{4.3.20}$$

here $\alpha_1 = 0$. The reaction degree can be verified to be correct through the flow coefficient

$$^{o}R = \frac{1}{2} + \phi\left(\frac{\tan\beta_2 + \tan\alpha_2}{2}\right) \tag{4.3.21}$$

A merit parameter is introduced to describe the boundary layer's level of adherence to the back of the rotor and stator blades profiles, the diffusion factor (D). It is derived through the relationship between circulation and lift, linking the relative radial velocities at the stage inlet and exit to the maximum lift coefficient expressible by the blades.

$$\begin{cases} D_{rot} = 1 - \frac{W_{2m}}{W_{1m}} + \frac{|W_{2m,\theta} - W_{1m,\theta}|}{2\sigma(r_m)W_{1m}} \\ D_{stat} = 1 - \frac{u_1}{u_{2m}} + \frac{|u_{3m,\theta} - u_{2m,\theta}|}{2\sigma(r_m)u_{2m}} \end{cases} \tag{4.3.22}$$

Where the $\sigma(r_m)$ (solidity) expresses the ratio between the blades' chord length and the relative distance, and it is a function of the radial coordinate (it is evaluated at the mean radius for consistency with the adopted strategy)

$$\sigma(r) = \frac{N_b c}{2\pi r} \tag{4.3.23}$$

a suitable value for Reynolds number is guessed to calculate the average chord length from the inverse formula of the Reynolds number

$$c_m = \frac{Re\nu}{W_{1m}} \tag{4.3.24}$$

Now, σ (which, for the fan is taken to be the same for both stator and rotor) is hypothesised. This is used to obtain the D factor.[2]

4.3.1 Blade sizing results for fan

Symbol	Description	Value	Unit
$r_{tip,1}$	Tip radius	1.9083	$[m]$
$r_{hub,1}$	Hub radius	0.5725	$[m]$
$\left(\frac{r_{hub}}{r_{tip}}\right)_1$	Radius ratio	0.3000	$[-]$
r_m	Mean radius	1.2404	$[m]$
$r_{tip,2}$	Tip radius	1.7547	$[m]$
$r_{hub,2}$	Hub radius	0.7261	$[m]$
$\left(\frac{r_{hub}}{r_{tip}}\right)_2$	Radius ratio	0.4138	$[-]$
^oR	Degree of reaction	0.8691	$[-]$
ϕ	Loading coefficient	0.5930	$[-]$
ψ	Flow coefficient	0.2681	$[-]$
α_1	Absolute velocity angle	0	$[^o]$
α_2	Absolute velocity angle	23.82	$[^o]$
β_1	Relative velocity angle	59.33	$[^o]$
β_2	Relative velocity angle	51.22	$[^o]$
$\frac{W_{1m}}{W_{2m}}$	Speed ratio for De Haller comparison	0.82	$[-]$
$\frac{u_1}{u_{2m}}$	Speed ratio for De Haller comparison	0.91	$[-]$
$\sigma(r_m)$	Solidity	0.5544	$[-]$
D_{rot}	Diffusion factor rotor	0.3886	$[-]$
D_{stat}	Diffusion factor stator	0.4494	$[-]$

Table 2: Fan blade sizing

The obtained value for sigma is less than 0.7, in line with the isolated foil theory. Also, the D factor, which theoretically should be less than 0.55, is acceptable.

4.4 Materials and manufacturing technologies

Little data is available on the materials used to make the GE9X, but there are numerous improvements over its predecessors due to the extensive use of 3D printing and ceramic matrix composites. The materials and construction techniques involved in this engine are discussed here.

The fan is made of latest-generation carbon fiber, which is very strong but lighter than titanium. The leading edge of the fan, which is the part most exposed to both mechanical stress and harsh atmospheric conditions (rain, humidity, contaminants), is covered with a steel alloy. This alloy is known for its robustness, mechanical load resistance, good corrosion resistance, and beneficial thermal properties.

The trailing edge is covered with fiberglass, providing a good balance between lightness and strength, and it can be easily molded to meet aerodynamic requirements. The fan case is made using 3D woven composite carbon fiber through a resin transfer molding (RTM) process. RTM is a low-pressure molding process where a mixture of resin and catalyst is injected into a closed mold containing the fiber. Once the resin has hardened, the mold can be opened, and the finished component can be removed.

5 Compressors

Figure 5: Safran Aero Boosters[3] low pressure compressor

The GE9X engine's fan and compressors are marvels of modern engineering, showcasing significant advancements in aerodynamics and material science. The compressor system is divided into two main sections: the low-pressure compressor (LPC) and the high-pressure compressor (HPC). The LPC consists of three stages, while the HPC comprises eleven stages, achieving a remarkable overall pressure ratio. The low-pressure compressor is coupled to the low-pressure turbine, whereas the high-pressure compressor is coupled to the high-pressure turbine. The coupling mechanism ensures that the power generated by the turbine is utilized to drive the compressors.

The use of 3D printing and ceramic matrix composites in both fan and compressor components has led to significant improvements over previous engine models, including reduced weight, increased durability, and better performance. These innovations contribute to the GE9X's overall efficiency, reducing fuel consumption and lowering emissions.

5.1 Analytic steps for Compressors resolution

Same compression ratio is taken to be feasible for each stage. Accordingly β for low pressure compressor and high pressure compressor can be seen proportional to the stage count.

$$\beta = \beta_c^{\frac{n_{\text{stage}}}{\text{total stages}}} \tag{5.1.1}$$

Air flow rate through compressors is related directly to the bypass ratio, BPR

$$\dot{m}_{a,h} = \frac{\dot{m}_a}{1 + BPR} \tag{5.1.2}$$

[3]GE90, GP7200, GEnx-1B/2B and GE9X are equipped with compressors manufactured by Safran Aero Boosters

Supposing polytropic compression, the total temperature of air leaving the compressors is

$$T_{0,out,s} = T_{0,in} \left(\beta\right)^{\frac{\gamma-1}{\gamma}} \tag{5.1.3}$$

Using polytropic efficiency, η the real temperature can be easily found

$$T_{0,out} = T_{0,in} + \frac{\left(T_{0,out,s} - T_{0,in}\right)}{\eta} \tag{5.1.4}$$

While the pressure is simply the pressure ratio, β, times the pressure entering the compressors

$$P_{0,out} = \beta P_{0,in} \tag{5.1.5}$$

Finally power required to run the compressor stages is

$$PWR = \frac{1}{\eta_m} c_p \dot{m}_{a,h} \left(T_{0,out} - T_{0,in}\right) \tag{5.1.6}$$

5.2 Sizing using Velocity triangles

In both low-pressure and high-pressure compressors, a slightly higher hub to tip ratio is assumed, typically around 0.5. Following the same steps as done for the fan, this procedure is applied.
The main difference lies in the number of stages. While the fan has only one stage, the compressor consists of multiple stages. In this analysis, it is assumed that thermodynamic variables (like Temperature, Pressure, and Enthalpy) have a consistent change within each stage. For compressors, the exact number of blades in each stage is unknown. So, a concept called solidity is introduced to calculate the blade count. This value will be rounded to the nearest whole number, adjusting the solidity factor previously set. In the low-pressure compressor, the rotor is more likely to experience separation issues. Hence, a higher solidity factor is chosen to ensure a more even effect within the stage. Conversely, in the high-pressure compressor, the opposite is true.

5.2.1 Blade sizing results for compressors

The following table shows sizing parameters for both low and high pressure compressors.

Symbol	Description	Low Pressure (1st stage)	High Pressure (1st stage)	Unit
$r_{tip,1}$	Tip radius	0.5587	0.4303	$[m]$
$r_{hub,1}$	Hub radius	0.2794	0.2151	$[m]$
$\left(\frac{r_{hub}}{r_{tip}}\right)_1$	Radius ratio	0.5000	0.5000	$[-]$
r_m	Mean radius	0.4190	0.3227	$[m]$
$r_{tip,2}$	Tip radius	0.5322	0.4045	$[m]$
$r_{hub,2}$	Hub radius	0.3059	0.2409	$[m]$
$\left(\frac{r_{hub}}{r_{tip}}\right)_2$	Radius ratio	0.5748	0.5955	$[-]$
^oR	Degree of reaction	0.8055	0.8619	$[-]$
ϕ	Loading coefficient	0.8011	0.5146	$[-]$
ψ	Flow coefficient	0.3889	0.2763	$[-]$
ΔT_0	Temperature step per stage	22.49	37.23	$[K]$
α_1	Absolute velocity angle	0	0	$[^o]$
α_2	Absolute velocity angle	25.90	28.23	$[^o]$
β_1	Relative velocity angle	51.30	62.77	$[^o]$
β_2	Relative velocity angle	37.33	54.59	$[^o]$
$\frac{W_{1m}}{W_{2m}}$	Speed ratio for De Haller	0.79	0.79	$[-]$
$\frac{u_1}{u_{2m}}$	Speed ratio for De Haller	0.90	0.88	$[-]$
$\sigma_r(r_m)$	Rotor's solidity	1.2027	1.0124	$[-]$
$\sigma_s(r_m)$	Stator's solidity	1.0000	1.2000	$[-]$
D_{rot}	Diffusion factor rotor	0.3398	0.3059	$[-]$
D_{stat}	Diffusion factor stator	0.3188	0.3160	$[-]$
N_b	Blade count	28	36	$[-]$

Table 3: Compressors blades sizing

5.3 Materials

Each stage's compressor blades are individually affixed to the disc or drum using spacers, fasteners, and bolts. The disc comprises disks and rotor blades, with each rotating stage comprising numerous individual components. The initial five stages of the GE9X engine's high-pressure compressor are classified as *blisks* (bladed disks). A blisk constitutes a single part integrating the disk and the rotating blades through additive manufacturing.

By eliminating the necessity to attach the blades to the disk, the number of compressor components is reduced. The blades' integration significantly diminishes drag and augments the overall efficiency of the GE9X engine by up to 8%. Moreover, this design eradicates the primary source of crack initiation and propagation in blade dovetail slots (attachments) [3].

6 Combustion Chamber

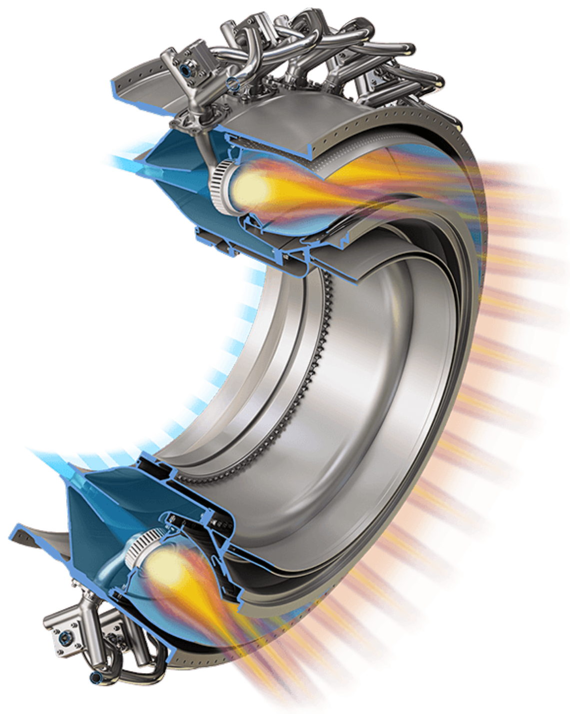

Figure 6: GE9X Combustion chamber

The primary role of the combustion chamber is to ignite a substantial amount of fuel via spray nozzles. This ignition occurs through interaction with large volumes of air supplied by the compressor. The released heat expands and accelerates the fluid uniformly, resulting in a continuous flow of gas.
Despite the confined space, it's crucial for this process to happen with minimal pressure loss and maximum thermal efficiency. Additionally, the burner must sustain stable and efficient combustion across all operating conditions. In this segment, the combustion process is discussed, from fuel nozzle tips to the cooling system designed to reduce gas temperature before reaching the turbine.

6.1 General description

The combustion chambers of the GE9X and GE90, both *high-bypass ratio* turbofan aircraft engines manufactured by GE Aviation for the Boeing 777 series, showcase notable differences in design and materials, reflecting advancements in engine technology.

GE9X combustion chamber incorporates cutting-edge technologies aimed at enhancing efficiency and reducing emissions. One key feature is the **Twin Annular Pre-mixing Swirler (TAPS)**, which premixes fuel with air before combustion, leading to lower NOx emissions compared to traditional combustors, as it will be discussed later [4].

Additionally, the use of 3D additive manufactured fuel nozzle tips, a novel combustor dome design, and lightweight yet durable ceramic matrix composite (CMC) inner and outer liners contribute to improve performances. These CMC liners not only enhance durability but also require less cooling air, facilitating a more efficient lean-burn combustion process.

Details about the combustion chamber of the GE9X are not readily available. However, considering the advancements showcased in the GE90, it can be deduced that the GE9X may have a better level of advanced materials and design features.

So, while both engines boast high-efficiency combustion chambers, the GE9X stands out with its innovative technologies and materials, resulting in up to 10% greater fuel efficiency compared to its predecessor, the GE90.

6.2 Analytic steps for Combustion Chamber resolution

Within the combustion chamber of this propulsion system, the airflow from the conduit connecting previous stages, namely the hot mass flow rate, enters, along with the conduit from the tanks, namely the fuel mass flow rate. Exiting the combustion chamber is the conduit leading to the subsequent stages of this propulsion system, namely the mass flow rate of combusted gases, which will be the sum of the mass flow rates of fuel and air from the hot conduit. So, the mass balance is as follows:

$$\dot{m}_{eg} = \dot{m}_{a,h} + \dot{m}_{a,c} \tag{6.2.1}$$

Certainly, the equations of energy conservation can be developed, expressing it as the product of the mass flow rates and the enthalpy:

$$\dot{m}_{eg} h_{eg} = \dot{m}_{a,h} h_a + \dot{m}_{a,c}(h_{of} + Q\eta_{cc}) \tag{6.2.2}$$

Combining mass conservation in the first equation with energy conservation in the second equation, and assuming that the term h_{of} is negligible compared to the term $(Q\eta_{cc})$, and also assuming that the kinetic part of enthalpy is negligible, and assigning the maximum temperature at the exit of the combustion chamber as the maximum tolerable temperature by the turbines in the subsequent stage, the approximation $h_0 = c_p T_0$ is obtained. The following steps can be written to find the definition of the fuel/air mass ratio f that was used:

$$\dot{m}_{eg} c_{p,eg} T_{max} = \dot{m}_{a,h} c_{p,a} T_{0,in} + \dot{m}_{a,c} Q\eta_{cc} \tag{6.2.3}$$

$$(1 + \frac{\dot{m}_{a,c}}{\dot{m}_a}) c_{p,eg} T_{max} = c_{p,a} T_{0,in} + \frac{\dot{m}_f}{\dot{m}_a} Q \cdot \eta_{cc} \tag{6.2.4}$$

$$(1 + f) c_{p,eg} T_{max} = c_{p,a} T_{0,in} + f Q\eta_{cc} \tag{6.2.5}$$

And finally

$$f = \frac{-c_{p,a} T_{0,in} + c_{p,eg} T_{max}}{\eta_{cc} Q - c_{p,eg} T_{max}} \tag{6.2.6}$$

Once f is found, the calculations for this stage can be concluded by determining the fuel mass flow rate and the outlet pressure from the combustion chamber.

$$\dot{m}_f = f \dot{m}_{a,h} \tag{6.2.7}$$

$$P_{0,out} = \Pi_{cc} P_{0,in} \tag{6.2.8}$$

6.3 TAPS

TAPS functions by pre-mixing fuel with air in two concentric rings before combustion. This pre-mixing process occurs through atomizing nozzles that create a finely atomized mixture of air and fuel. The result is a more uniform and stable flame that burns at lower temperatures compared to traditional combustors. This reduces the formation of nitrogen oxides (NOx), a key pollutant.

It features two annular fuel-air swirlers adjacent to nested fuel nozzles. The swirl creates a more homogeneous and leaner mix of fuel and air, which burns at lower temperatures than in previous jet engine designs. The lower temperatures generated in the TAPS combustor results in significantly lower NOx levels [4].
Tests have shown that at comparative thrust levels, GE9X NOx emissions will be more than 30% lower than the NOx emissions of GE's highly popular CF6 engines powering commercial Wide-Body aircrafts today.

The GE9X emissions goal at entry into service is to be about 50% below the latest NOx limits established in 2008.

6.4 Fuels

One of the biggest advantages of this engine is the fact that it does not need a specific fuel to work, leading to an high-level *versatility* in the civil aviation field.
Said that, this section will briefly talk about the principal properties the most common civil aviation fuels. In particular the following fuels are discussed:

- Jet-A
- Jet-A1
- Jet-B

6.4.1 Jet-A

Jet A is a kerosene grade fuel specifically designed for use in turbine engine aircraft, including jet aircraft and turboprops.
Jet-A is similar to diesel fuel but has a **higher flash point** and **lower freezing point** to ensure safe and reliable operation in a wide range of temperatures and altitudes.
One of the key characteristics of Jet-A fuel is its **high energy density**, which allows aircraft to achieve long-range flights with efficient fuel consumption. It also meets strict specifications for purity and performance set by aviation authorities such as ASTM International and the International Air Transport Association (IATA).
Jet-A fuel is typically stored and distributed in large fuel storage facilities at airports and is delivered to aircraft via fuel trucks or hydrant systems. Airlines closely monitor fuel quality and quantity to ensure the safety and efficiency of their flights.

6.4.2 Jet-A1

Jet-A1 is a type of aviation fuel commonly used in commercial and military aircraft worldwide. It is very similar to Jet-A fuel, but with an even **lower freezing point**, making it suitable for use in colder climates and at higher altitudes.

Just like Jet-A, Jet-A1 is a refined kerosene-based fuel specifically formulated for use in turbine engine aircraft, including jet aircraft and turboprops. It meets stringent specifications for purity and performance set by organizations.
Jet-A1 fuel shares many characteristics with Jet-A, including its high energy density, which enables aircraft to achieve long-range flights with efficient fuel consumption.

Jet-A1 combustion simulation

It is possible, using the software NASA-CEA, to simulate combustion reactions and evaluate specific properties of the mixture.
This software was used to simulate the combustion of the Jet-A1 fuel to evaluate the adiabatic flame temperature and the specific heat of gases after the combustion process. It was also analyzed how temperature change at different values of O/F ratio.
The results can be seen in the following graph, figure 7:

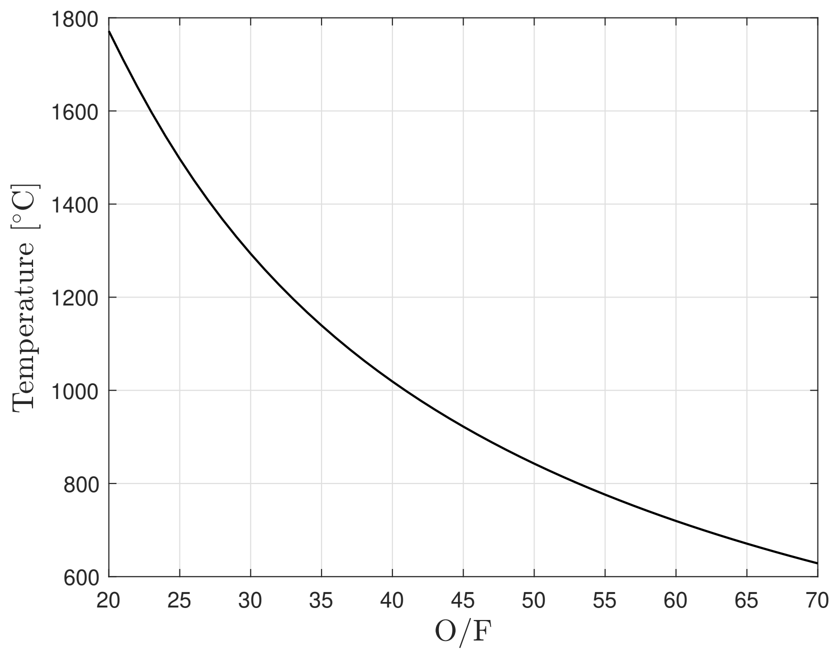

Figure 7: Adiabatic flame temperature at different O/F

It is clear that a lower O/F ratio generates a higher flame temperature, due to an higher quantity of fuel. On the other hand if there's too much fuel and no oxidant the flame temperature starts decreasing.
It is also possible to observe the deviation of a given Temperature value at a given O/F by looking at the deviation between the current value and the previous values, as shown in this graph, figure 8.

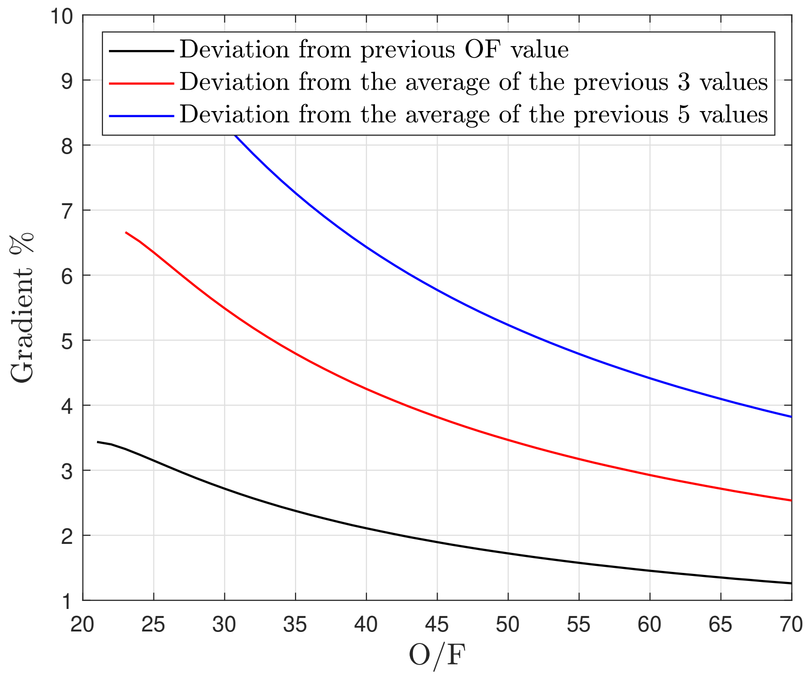

Figure 8: Gradient vs O/F

An O/F value that maximise the flame temperature and the fuel combustion is needed and at the same time there is the need to minimize the average costs. Then a reasonable O/F value could be $40 < O/F < 60$ because in this region the derivatives of the three curves are lower so there is a less decrease in performance if the O/F shift value.
This is also a good compromise in terms of performance's stability.

6.4.3 Jet-B

Jet-B is a type of aviation fuel primarily used in North America, although its usage has declined in recent years in favor of Jet-A and Jet-A1 fuels. It is a high-flash-point kerosene-based fuel with a relatively **low freezing point**, making it suitable for use in colder temperatures.
One of the distinguishing characteristics of Jet-B fuel is its higher volatility compared to Jet-A and Jet-A1. It contains a higher proportion of lighter hydrocarbons, which gives it a higher vapor pressure and makes it more prone to vapor lock in aircraft fuel systems. For this reason, Jet-B fuel is less commonly used than Jet-A and Jet-A1. Matter of fact, the consequences of an higher flash point is an higher flammability which is good for the combustion efficiency and performance but absolutely not for safety.
However, Jet-B still finds applications in certain military aircraft and in some remote areas where Jet-A or Jet-A1 may not be readily available.

6.5 Materials

GE-9X combustion chamber is made of highly performing materials designed specifically with new technologies like additive manufacturing. The two main materials that compose the combustion chamber are:

- Nickel-based superalloys
- Ceramic Matrix Composite (CMC)

6.5.1 Nickel-based superalloys

Nickel-based superalloys are a class of advanced materials primarily composed of nickel, with significant additions of other elements such as chromium, cobalt, iron, aluminum, titanium, and others. These alloys are used because of their exceptional mechanical strength, resistance to corrosion, oxidation, and thermal stability, even at elevated temperatures.

Resistance to corrosion and oxidation is a key factor in a combustion chamber. Nickel-based superalloys are highly resistant both to oxidation and corrosion, even in aggressive environments such as high-temperature combustion gases and corrosive atmospheres. This resistance helps to:

- Extend the service life of components
- Reduce maintenance requirements

These are both key factor in a combustion chamber design.

There's another important factor regarding Nickel-based superalloys: the fact that *they can be built with additive manufacturing.*

Additive manufacturing offers several advantages such as:

1. Complex geometries made easier
2. Reduced material Waste [4]
3. Easier customization and rapid prototyping

While additive manufacturing offers numerous benefits, there are challenges associated with producing components from nickel-based superalloys. These include controlling microstructure and porosity, managing residual stresses, and ensuring dimensional accuracy.

6.5.2 Ceramic Matrix Composite (CMC)

Ceramic matrix composites (CMC) are a critical pathway on GE Aviation's technology roadmap.

Tests on a GE-9X demonstrator engine with GE-9X CMC components revealed the importance that this ultra-lightweight, heat-resistant material have on global engine performance.

The GE-9X CMC demonstrator engine accumulated 2,800 endurance cycles at a GE's test cell in Peebles, Ohio and an Avio Aero's test cell in Naples, Italy.

CMC materials are a great choice because of these two main reasons:

1. They have **one-third** the density of metal alloys, so they reduce engine's overall weight and drastically improve fuel efficiency

[4] Important factor for green economy

2. They have great high-temperature properties, greatly enhancing engine performance, durability, and fuel economy. Since CMCs are far more heat resistant than metal alloys, they require less cooling air in the engine's hot section. This air instead can be used in the engine flow path, enabling it to run more efficiently.

The GE9X engine for the Boeing 777X aircraft will incorporate CMCs in the inner and outer combustor liners, HPT stage 1 shrouds and stage 1 and stage 2 nozzles.

6.6 Cooling systems

GE9X engine generates a significant amount of heat during operation, particularly in the combustion chamber and hot sections of the engine. To prevent overheating and ensure optimal performance and durability, not just one but various cooling systems are employed throughout the engine.

- *Internal cooling passages*: the engine components, such as turbine blades and combustors, have internal passages through which cooling air flows. This cooling air absorbs heat from the hot components and is then directed out of the engine or sometimes reused for other purposes.

- *Film cooling*: thin films of air are often directed over the surfaces of hot components to create a protective barrier that shields them from the high temperatures of combustion gases.

- *Effusion cooling*: this is kind of a mix of the previous method, this method involves passing cooling air through a series of small holes on the surface of components, such as turbine blades, to create a cooling film that helps dissipate heat.

- *Ceramic coatings*: some engine components may be coated with special ceramic materials that provide thermal insulation and protection against high temperatures, they are discussed in the material section.

- *Secondary specific airflow*: in addition to cooling air used for combustion and component cooling, secondary airflow may be employed specifically for cooling purposes in certain areas of the engine.

- *Fan air*: in high-bypass turbofan engines like the GE9X, a significant portion of the airflow bypasses the core of the engine and is instead directed around the engine core. This bypass air is designed to help cooling out various engine components.

7 Turbines

Figure 9: GE9X LPT

The turbine is a prime mover turbomachinery that facilitates the exchange of work with the fluid. In comparison to the compressor, the function of the turbine is simplified by the negative pressure gradient (along the axis of the turbine itself), which significantly mitigates boundary layer separation issues. However, turbines are subjected to substantial thermal stresses, compounded by high rotational speeds, which complicates their construction. Cooling of the turbine blades is crucial to maximize the impact of fluid enthalpy entering the turbomachinery (and exiting the combustion chamber). The function of the turbine is diametrically opposed to that of the compressor, a distinction reflected in the sequence of turbine blade arrangement, where the stator precedes the rotor.
The thermal load on the turbine is a critical factor not to be underestimated. The mechanical limit of the maximum temperature manageable by the turbine dictates the operational limit of the combustion chamber, thereby enabling a greater enthalpy differential available for expansion.

7.1 Analytic steps for High Pressure Turbine resolution

Incoming exhaust gases and total mass flow rates are worked out through the fuel ratio as follows

$$\dot{m}_{\text{eg}} = f\,\dot{m}_{a,h} \tag{7.1.1}$$

$$\dot{m}_{a} = \dot{m}_{a,h} + \dot{m}_{\text{eg}} \tag{7.1.2}$$

HPT acts as the power source for the HPC. As such, the power consumption of the HPC is factored into the calculation of the true total temperature following the final stage of the HPT

$$T_{0,\text{out}} = -\frac{\text{PWR}_{\text{hpc}} - T_{0,\text{in}}\,c_{p,\text{eg}}\,\eta_{\text{hpt},m}\,\dot{m}_{a,h}\,(f+1)}{c_{p,\text{eg}}\,\eta_{\text{hpt},m}\,\dot{m}_{a,h}\,(f+1)} \tag{7.1.3}$$

Also, same stage isentropic temperature is found in order to retrieve LPT input total pressure

$$T_{0,\text{out},s} = T_{0,\text{in}} - \frac{T_{0,\text{in}} - T_{0,\text{out}}}{\eta_{\text{hpt}}} \tag{7.1.4}$$

$$P_{0,\text{out}} = \frac{T_{0,\text{in}}}{\left(\frac{T_{0,\text{in}}}{T_{0,\text{out},s}}\right)^{\frac{\gamma_{\text{eg}}}{\gamma_{\text{eg}}-1}}} \tag{7.1.5}$$

7.2 Analytic steps for Low Pressure Turbine resolution

The logical procedure applied to the Low-Pressure Turbine (LPT) closely mirrors the one previously outlined, albeit with a notable distinction: this component must act as a power supplier for both the Fan and the High-Pressure Compressor (HPC) unit

$$\dot{m}_{\text{eg}} = f\,\dot{m}_{a,h} \tag{7.2.1}$$

$$\dot{m}_a = \dot{m}_{a,h} + \dot{m}_{\text{eg}} \tag{7.2.2}$$

Therefore, when determining the actual enthalpy leap required, the power consumption of both components must be aggregated. Naturally, this calculation accounts for the efficiencies inherent to proper component.

$$T_{0,\text{out}} = -\frac{\text{PWR}_f + \text{PWR}_{\text{lpc}} - T_{0,\text{in}}\, c_{p,\text{eg}}\, \eta_{\text{lpt},m}\, \dot{m}_{a,h}\ (f+1)}{c_{p,\text{eg}}\, \eta_{\text{lpt},m}\, \dot{m}_{a,h}\ (f+1)} \tag{7.2.3}$$

$$T_{0,\text{out},s} = T_{0,\text{in}} - \frac{T_{0,\text{in}} - T_{0,\text{out}}}{\eta_{\text{lpt}}} \tag{7.2.4}$$

$$P_{0,\text{out}} = \frac{T_{0,\text{in}}}{\left(\frac{T_{0,\text{in}}}{T_{0,\text{out},s}}\right)^{\frac{\gamma_{\text{eg}}}{\gamma_{\text{eg}}-1}}} \tag{7.2.5}$$

7.3 Blade stress and temperature distribution qualitative analysis

Turbine blades experience various types of stress, including torsional stress at the root due to twisting moments generated by high rotation speeds, and mechanical stress at the apex caused by high centrifugal forces. The intermediate area undergoes less mechanical stress. Consequently, the project will focus on optimizing the distribution of maximum temperature along the centerline of the blades, aiming for lower temperatures in the highly stressed root and apex areas.

To achieve this, careful planning of temperature distribution from the combustion chamber is necessary, particularly in the dilution chamber, a complex process. The pattern factor serves as a key parameter for estimating outlet temperature distribution, offering an initial approximation. The profile factor, which depends on the maximum average circumferential temperature, describes the radial temperature distribution, with emphasis on the average radial temperature distribution at the combustion chamber exit section.

In modern engines, the average radial temperature distribution at the exit section of the combustion chamber exhibits a non-uniform peak above 50% of the blade. Therefore, defining the turbine profile factor is more appropriate. Additionally, it is crucial to avoid pressure oscillations in the combustion chamber, as they can lead to instability phenomena.

7.4 Materials

Ceramic matrix composites (CMCs) are utilized in specific components of the turbine system to withstand elevated temperatures. These include the first-stage HP turbine shroud, the first and second stage HP turbine nozzles, and the inner and outer combustor linings. These CMC components, which are static components, operate at temperatures approximately 500 °F (260 °C) hotter than traditional nickel alloys, albeit with some cooling. CMCs offer enhanced strength at only one-third of the weight compared to metals, necessitating significantly less cooling, approximately 59% less, while maintaining structural integrity.

The first-stage turbine blades, however, are not constructed from CMCs due to the extreme heat and centrifugal forces they must withstand. Improvements in this area are anticipated in future advancements of turbine technology.

Additionally, while the high-pressure turbine is crafted from powdered metal, the low-pressure turbine airfoils are fabricated from titanium aluminide (TiAl). TiAl components offer superior strength, reduced weight, and increased durability in comparison to their nickel-based counterparts.

7.5 Blade Cooling Systems and Manufacturing Hints

In the pursuit of higher thermal efficiency, turbine engines rely on elevated turbine inlet temperatures. However, these high temperatures pose a risk of damage to turbine blades due to significant centrifugal forces and material weaknesses. To mitigate this risk, effective turbine blade cooling systems are indispensable. Cooling methods typically involve either air or liquid coolant. Air cooling, achieved by bleeding air from the compressor, creates a protective air cushion that is blown through blade holes, thermally isolating the blade. This method, consuming only 1-3% of the main airflow, can effectively reduce temperatures by 200-300 °C. Air may flow from the root to the blade tip, driven by pressure differentials and centrifugal forces, facilitates convective heat extraction. Early measures also included creating small holes on the blade surface near the leading edge to channel air over the airfoil, forming a thin protective layer known as "film cooling" to reduce heat exchange with the mainstream flow and manage the blade's boundary layer. If both techniques are implemented simultaneously, the coil inside the blade must be designed to enhance heat exchange with the air and thus its cooling. Traditional machining methods are inadequate for producing this complex structure (see figure 10); instead, additive manufacturing processes such as three-dimensional printing are utilized.

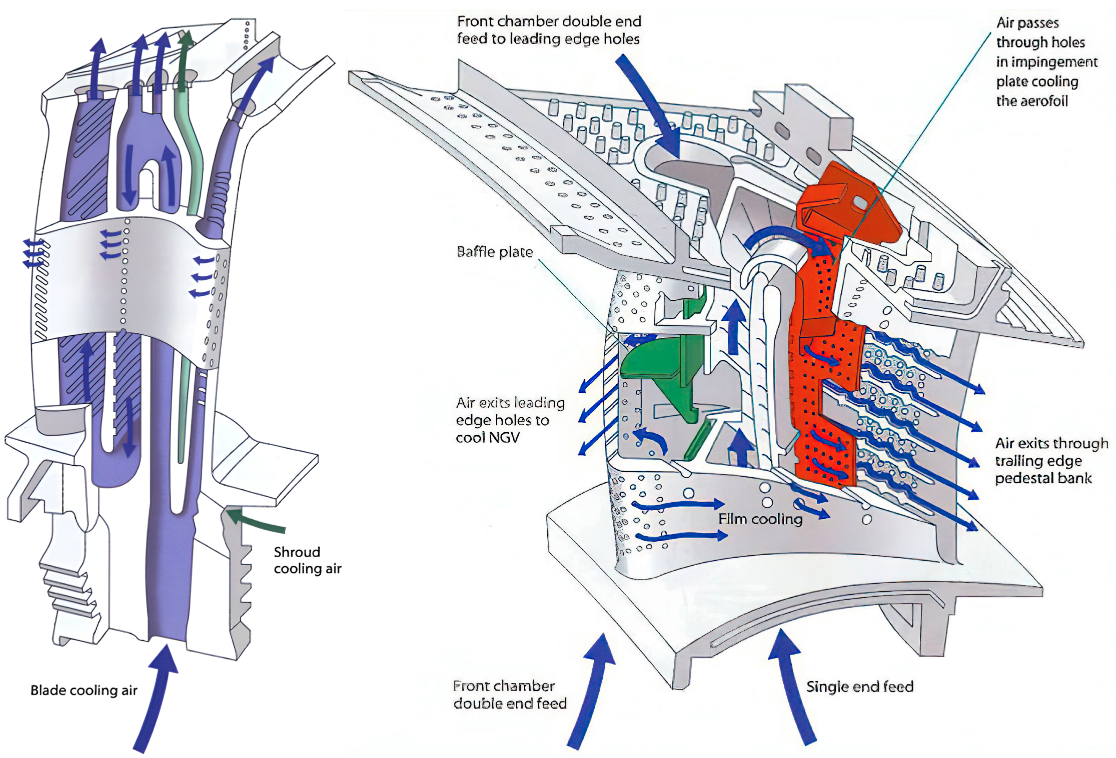

Figure 10: Turbine blade cooling technology[5]

8 Nozzles

Figure 11: The GE9X FETT[5] during its first test cycle in early 2016

Efficient propulsion system design is crucial for modern aircraft to achieve optimal performance and fuel efficiency. Among the key components of the GE9X, nozzles play a vital role in controlling exhaust gas flow and maximizing thrust generation. Understanding the expansion behavior of nozzles is essential for engine optimization, particularly during cruise conditions where performance demands are stringent.

8.1 Analytic steps for Nozzle resolution

The study utilizes governing equations to analyze gas behavior under conditions closely resembling real-world scenarios. Specifically, the research considers absence of heat exchange, constant altitude, and horizontal flight, typical of cruise conditions. The Boeing 777X, like its predecessors, employs step climbing during cruise rather than immediate altitude attainment, necessitating a study of nozzle behavior at various altitudes. Even though highly unrealistic altitudes from FL50 to cruise, FL390, were studied. A hypothesis was made, the nozzles expands air, from its total pressure and temperature to its static ambient conditions, or at least they try.

$$P_{out} = P_{amb(FL390)} \tag{8.1.1}$$

Due to nozzles being simply convergent it is impossible that the flow is accelerated over the speed of sound, owing to the sonic blockage immediately outside the nozzle. So the expansion can reach only the critical value

$$P_{out} < P_{crit} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} P_{0,in} \tag{8.1.2}$$

[5]First Engine To Test

This situation can be observed to be verified at all altitudes in both nozzles, supposed the speed is always $M = 0.85$, which is an unrealistic hypothesis adding to the assumption of constant throttle at all altitudes. This means the gases leaving the nozzles will always reach Mach one. But this does not imply that the actual exhaust speed is the same, due to other variable changes, as shown in figures 12 and 13.
If sonic conditions are found at the exhaust then the temperature will assume the critical values too

$$T_{out} = T_{0,in} \left(\frac{P_{out}}{P_{0,in}} \right)^{\frac{\gamma-1}{\gamma}} = T_{0,in} \frac{2}{\gamma + 1} \tag{8.1.3}$$

Now assuming perfect gas law the density ρ is calculated

$$\rho = \frac{P_{out}}{T_{out} R} \tag{8.1.4}$$

The exhaust speed, in presence of locked flow for both nozzles, is the speed of sound, so

$$u_{out} = a = \sqrt{\gamma R T_{out}} \tag{8.1.5}$$

finally the exhaust section of the nozzle is

$$A_{out} = \frac{\dot{m}}{\rho u_{out}} \tag{8.1.6}$$

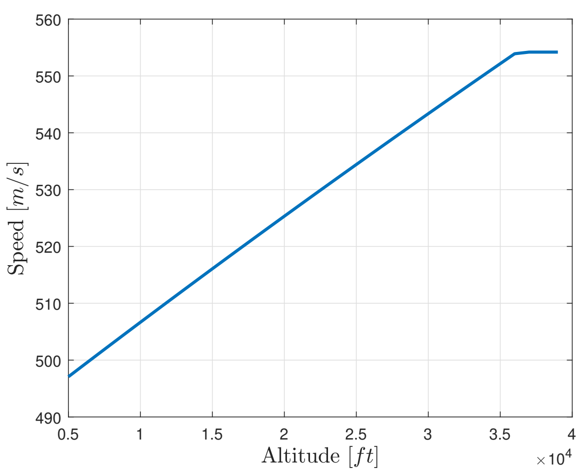

Figure 12: hot gas nozzle

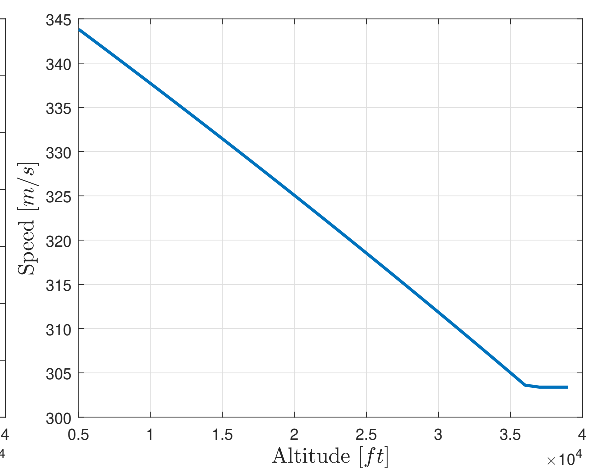

Figure 13: cold gas nozzle

Note that the cold gas cycle has no temperature increase with altitude due to gases not running through a combustion process. So it has decreasing temperature with altitude, which is followed by a speed chart going down too ,while the hot gas cycle has the opposite behaviour. The plateau in both speed plots are due to surpassing the 11,000 metres of altitude. Above this flight level the temperature at the inlet is the same and the pressure ratios in the nozzles between the inlet and exit area are the same too. This results in the same speed at nozzle exit even at different temperatures and pressures.

8.2 Convergent Nozzles benefits

Let's see why the engine has a convergent nozzle instead of a De Laval nozzle, even though it is never expanding the exhaust gas completely. First the gross thrust of a De Laval nozzle is calculated. To do this, the expansion is set to be the ambient pressure, for example at maximum cruise altitude. In this situation the NPR, nozzle pressure ratio is

$$NPR = \frac{P_{0,in}}{P_{amb(FL390)}} = \begin{cases} 2.511 & \text{for hot gas nozzle} \\ 2.223 & \text{for cold gas nozzle} \end{cases} \tag{8.2.1}$$

It is possible to chart the relation between the NPR and the ratio of gross thrust between the convergent and De Laval.[6]. By recalculating the gross thrusts in both nozzles in a generic situation, where NPR and γ are both parameters, the following ratio equation can be found

$$\frac{F_{DeLaval}}{F_{conv}} = \sqrt{\frac{1 - NPR^{-\frac{\gamma-1}{\gamma}}}{\frac{\gamma-1}{\gamma+1}}} \frac{\gamma}{\gamma + \left(1 - (\frac{\gamma+1}{2})^{\frac{\gamma}{\gamma-1}} NPR^{-1}\right)} \tag{8.2.2}$$

which translates into a plot as shown in figure 14.

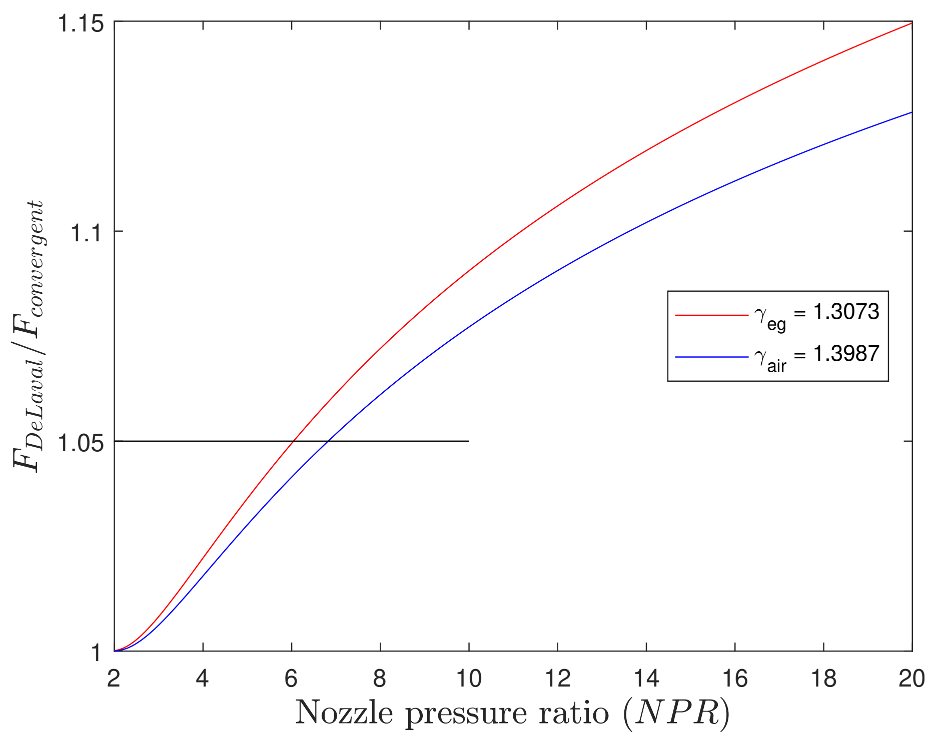

Figure 14: NPR vs gross thrust ratio

Now it is clear that to achieve a 5% gain in gross thrust using a De Laval nozzle a NPR equal to 6 is needed. That is clearly not our situation where at maximum the NPR is equal to 5.24, thus a convergent nozzle is enough due to almost no gross thrust benefits in using a De Laval nozzle. Furthermore making a simply convergent nozzle is easier than designing and creating a complex De Laval nozzle. It even comes with reduced weight and size.

8.3 Materials

The exhaust system of the GE9X is manufactured by Safran, a specialized company in the field. Safran Nacelles excels in exhaust system design and production, contributing to the aircraft's thrust by directing hot airflow from the engine while ensuring a continuous flow of cold air exiting the nacelle.
These systems are made from fine metallic alloys, typically ranging from 0.016 to 0.06 inches thick (0.4 to 1.5 mm), which reduces weight, offers corrosion resistance, and can withstand extreme temperatures exceeding 600°C (1112°F). To mitigate noise, honeycomb acoustic treatment is strategically applied to the GE9X exhaust system on the Boeing 777X. Thousands of holes are drilled into the composite skin to disperse noise, which is then captured within the honeycomb core. This process significantly diminishes engine noise, showcasing Safran Nacelles' unique expertise in engine environment acoustic technology. Furthermore adapting this solutions, rather than chevron nozzles, helps drastically reducing nozzle weight too.

9 Assessment of Results

9.1 Engine Thrust and Performance Parameters

GE9X engine has recorded a peek 597 kN takeoff thrust[7] while available thrust in cruise is unknown. So the only reference is the GE90, which provides a 70 kN thrust in cruise. **Available thrust** can be calculated as follows

$$F_{tot} = \dot{m}_{a,c}u_{cg} + \dot{m}_{eg}u_{hg} + A_{hg}P_{out,h} + A_{cg}P_{out,c} - (A_{cg} + A_{hg})(P_{amb,FL}) - \dot{m}_a u_{flight} \quad (9.1.1)$$

where the drags due to nacelles, inlet geometry and engine attack to wing are not counted in. Therefore the resulting thrust of 95.4 kN at FL390 has to be downgraded, according to the dismissed drag in calculations, for real life scenarios. This value, looking backwards to GE90 capabilities, is quite reasonable.
The **TSFC**, *Thrust specific fuel consumption*, is the fuel flow rate to available thrust ratio.

$$TSFC = \frac{\dot{m}_f}{F_{tot}} = \frac{\dot{m}_{a,h}f}{F_{tot}} \quad (9.1.2)$$

This parameter has to be minimized in a well performing engine in order to produce thrust with the minimum fuel expenditure.
The **Thermal efficiency** is the engine's capability to convert the thermal energy released from the fuel during a chemical reaction into a net gain of kinetic energy in the working medium.

$$\eta_{th} = \frac{(\dot{m}_{a,h} + \dot{m}_f)u_{eff,h}^2 + (\dot{m}_{a,c})u_{eff,c}^2 - \dot{m}_a u_{flight}^2}{2m_f Q} \quad (9.1.3)$$

where the effective speeds are expressed as

$$u_{eff,h} = \frac{F_{hg}}{m_{eg}} \ , \ u_{eff,c} = \frac{F_{cg}}{m_{a,c}} \quad (9.1.4)$$

The **Propulsive efficiency** is the propulsive power on jet power

$$\eta_{prop} = \frac{2F_{tot}u_{flight}}{m_{a,c}u_{eff,c}^2 + m_{eg}u_{eff,h}^2 - m_a u_{flight}^2} \quad (9.1.5)$$

Finally the **Global efficiency** is the product of the previous

$$\eta_{gl} = \eta_{th}\eta_{prop} \quad (9.1.6)$$

9.2 BPR analysis

The bypass ratio (BPR) of a turbofan engine is defined as the ratio of fan-to-core flow rate (mass air flow of the cold cycle over the one of the hot cycle)[6]:

$$BPR = \frac{\dot{m}_{a,c}}{\dot{m}_{a,h}} \quad (9.2.1)$$

For the GE9X, the BPR is set to 9.9. It's interesting to see how any change to its value affects the performance in order to understand if the chosen value turns out to be the optimal one or better results can be achieved. Thanks to the computing architecture created in MATLAB it's possible to loop the code for different values of BPR and note the trends of some relevant performance parameters, such as specific thrust, TSFC (Thrust Specific Fuel Consumption) and efficiency. To obtain these trends, formulas in the section 9.1 about Engine Thrust and Performance Parameters were used.

This analysis is referred to the fixed cruise altitude of $39000ft$ ($FL390$).
Regarding the following graphs, please note that the zero bypass ratio simulates a turbojet with the same inlet capture area, while the actual GE9X BPR is marked in black.

Thrust
In general, the magnitude of the thrust produced is directly proportional to the mass flow rates of the fluid flow through the engine. Then, it is logical to study thrust per unit mass flow rate as a figure of merit of a candidate propulsion system. In case of an air-breathing engine, the ratio of thrust to air mass flow rate is called specific thrust and is considered to be an engine performance parameter. The target for this parameter, is usually to be maximized, that is, to produce thrust with the least quantity of air flow rate, or equivalently to produce thrust with a minimum of engine frontal area. The specific thrust of the GE9X as a function of bypass ratio is shown in the following graph, figure 15.

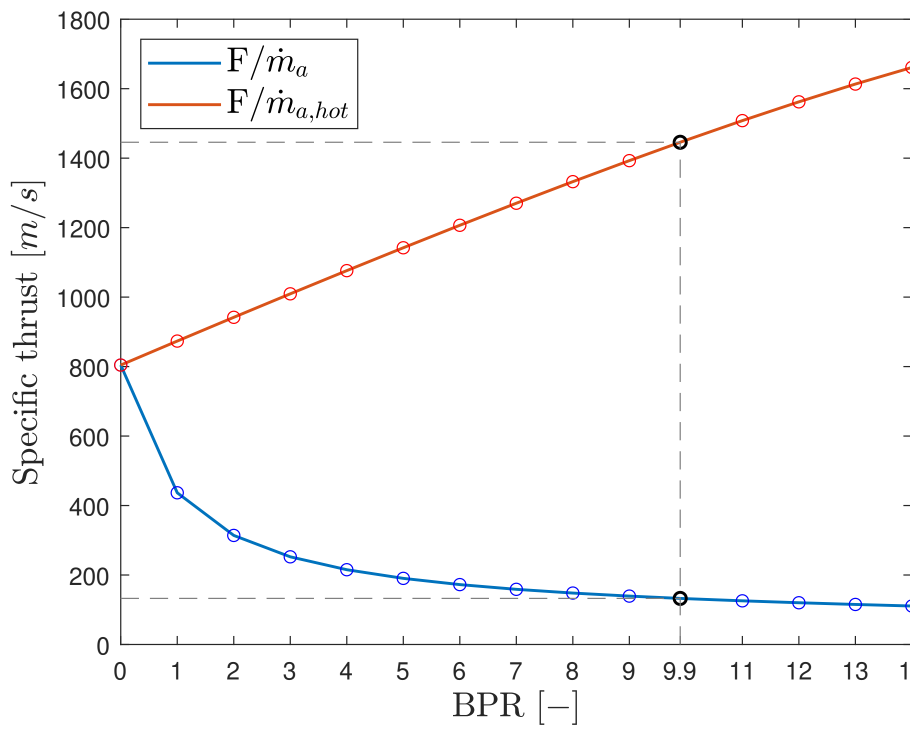

Figure 15: Specific thrust vs BPR (at FL390)

Since a turbofan produces its thrust by acting over a large mass flow of air, its specific thrust is lower than a turbojet (BPR = 0). It's clear how the engine thrust per unit inlet mass flow rate is reduced with increasing bypass ratio. However, the graph of the turbofan engine thrust per unit core mass flow rate (in red) shows that engine thrust continuously increases with bypass ratio.

Fuel consumption
The ability to produce thrust with a minimum of fuel expenditure is another parameter that is considered to be a performance parameter in an engine. In the commercial world, for example, the airline business, specific fuel consumption represents perhaps the most important parameter of the engine. For an airbreathing engine, the ratio of fuel flow rate per unit thrust force produced is called thrust-specific fuel consumption (TSFC). The target for this parameter is to be minimized, that is, to produce thrust with a minimum of fuel expenditure.

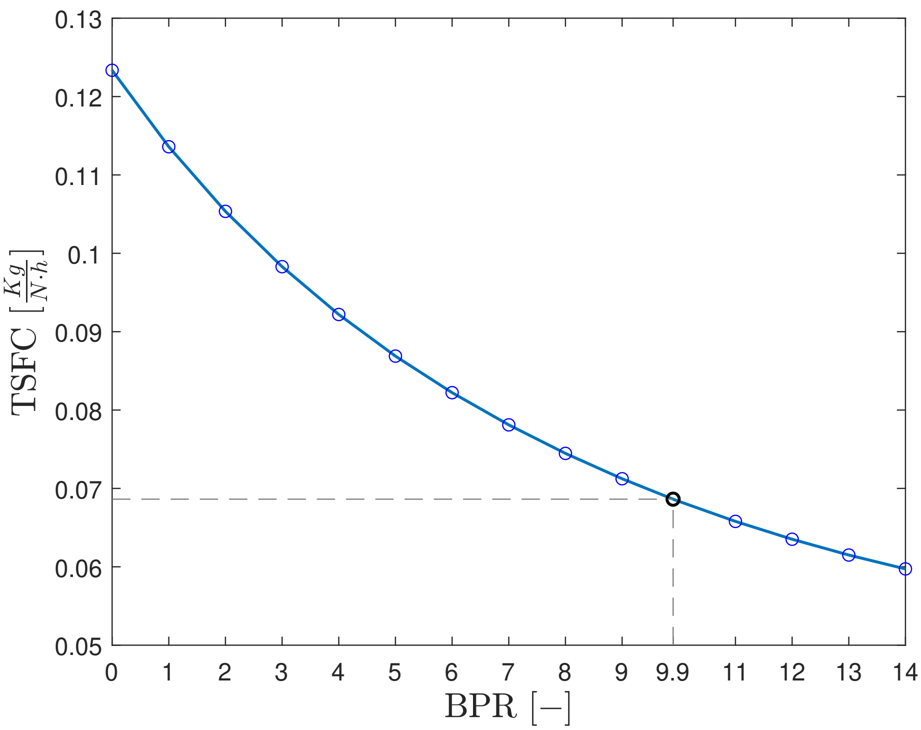

Figure 16: TSFC vs BPR (at FL390)

The plot, figure 16, shows continuous improvement with increasing bypass ratio. Indeed, the turbojet case of BPR $=0$ consumes the most fuel. The modern ultra-high bypass ratio turbofan engines in commercial aviation with a bypass ratio of $\sim$10 use $\sim$45% less fuel than a turbojet or $\sim$20% less fuel than a bypass ratio 5 turbofan engine.

Efficiency
The ability of an engine to convert the thermal energy inherent in the fuel (which is unleashed in a chemical reaction) to a net kinetic energy gain of the working medium is called the engine thermal efficiency.

The fraction of the net mechanical output of the engine which is converted into thrust power is called the propulsive efficiency. The principle behind the turbofan concept comes from sharing the power with a larger mass flow rate of air at a smaller velocity increment: the smaller the velocity increment across the engine, the higher the propulsive efficiency will be.

The product of the engine thermal and propulsive efficiency is called the engine overall, or global efficiency. The overall efficiency of an aircraft engine is therefore the fraction of the fuel thermal power that is converted into the thrust power of the aircraft.

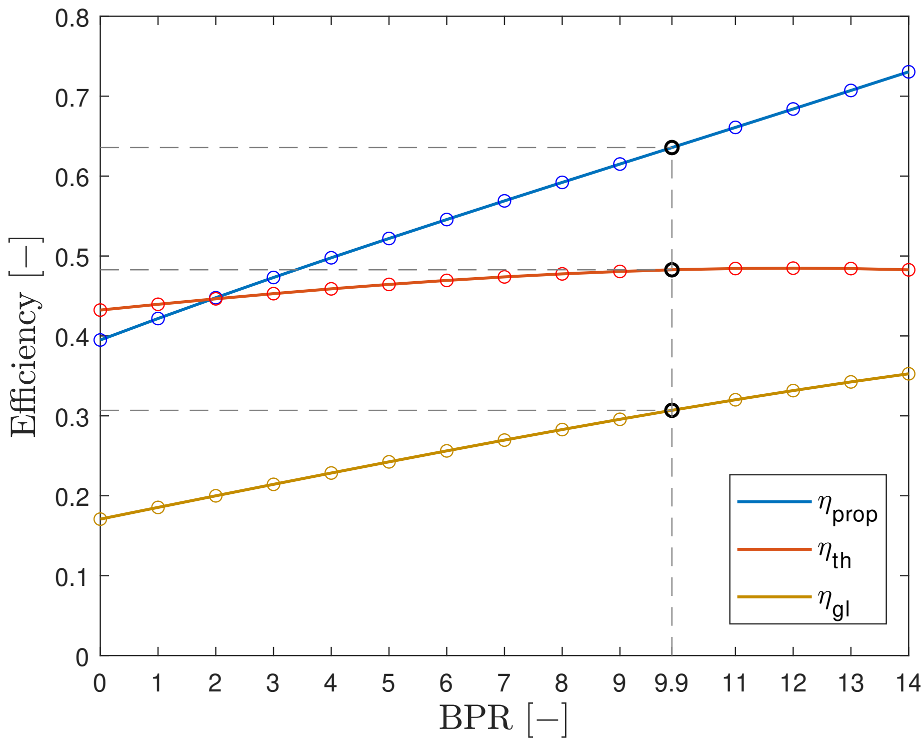

Figure 17: Efficiency vs BPR (at FL390)

The ideal thermal efficiency of a turbofan engine does not change with bypass ratio and, as seen, this point is borne out by data for the real GE9X turbofan engine as well, as shown in the graph, figure 17. The propulsive efficiency of a turbofan engine, in contrast to the thermal efficiency of such engines, is a strong function of the engine bypass ratio. Its value, which was the main target of improvement in a turbofan engine, shows continuous increase with bypass ratio. In fact, the lowest propulsive efficiency corresponds to the turbojet engine (BPR = 0), at ∼40% in this case. The overall efficiency improves with bypass ratio similar to the propulsive efficiency.

Conclusions

With a BPR of ∼10, the GE9X comes really close to be considered an Ultra-High Bypass (UHB) Turbofan Engine (this category typically starts from BPR ∼12) and this value is a great compromise for what concern both specific thrust and global efficiency, and it already guarantees a massive reduction in terms of fuel consumption.

More in general, as it was already noted, the specific thrust drops for a turbofan, in comparison to a turbojet engine that has zero bypass ratio, as more air is handled to produce the thrust in a bypass configuration. Therefore, this observation of lower specific thrust for a turbofan engine leads to the conclusion that bypass engines have, by necessity, larger frontal areas, again in comparison to a turbojet engine producing the same thrust.

So, the general conclusions are that for subsonic applications, the external drag penalty of a turbofan engine installation is significantly smaller than the benefits of higher engine propulsive efficiency.

As a result, in a subsonic cruise civil transport, for example, the trend has been and continues to be in developing higher bypass ratio engines, not only to save on fuel consumption (which still remains the main target) but to reduce noise and pollution as well.

9.3 Technological Advancements

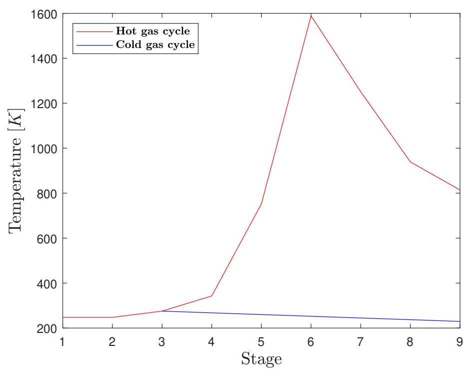

Figure 18: Temperature vs stage

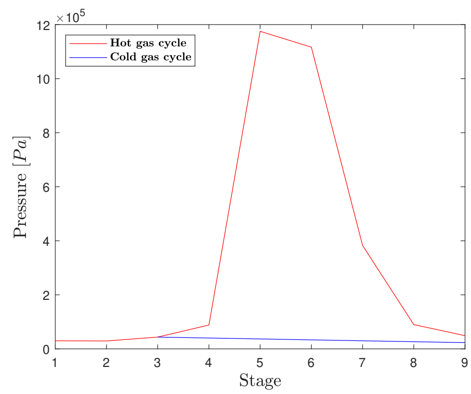

Figure 19: Pressure vs stage

Figures 18 and 19 depict the temperature and pressure variations across different stages of the engine for both the hot and cold gas cycles at FL390. The temperature graph (figure 18) shows a significant rise in temperature in the hot gas cycle, peaking in the middle stages before declining. This indicates the high thermal load handled by the engine. The pressure graph (figure 19) demonstrates the pressure distribution through the engine stages, with the hot gas cycle showing higher pressures compared to the cold gas cycle. These graphs highlight the engine's ability to manage thermal and pressure stresses effectively.

Symbol	Description	Value	Unit
F_{tot}	Total thrust	95423	$[N]$
F_{hot}	Hot Nozzle thrust	56263	$[N]$
F_{cold}	Cold Nozzle thrust	21929	$[N]$
$TSFC$	Thrust Specific Fuel Consumption	0.0686	$\left[\frac{kg}{N \cdot h}\right]$
η_{th}	Thermal efficiency	0.4827	$[-]$
η_{prop}	Propulsive efficiency	0.6357	$[-]$
η_{gl}	Global efficiency	0.3069	$[-]$
f	Fuel to air ratio	0.0276	$[-]$
$\dot{m}_a$	Airflow rate	719.43	$[\frac{kg}{s}]$
$\dot{m}_{a,h}$	Hot-Airflow rate	66.00	$[\frac{kg}{s}]$
$\dot{m}_{a,c}$	Cold-Airflow rate	653.43	$[\frac{kg}{s}]$
$\dot{m}_f$	Fuel flow rate	1.82	$[\frac{kg}{s}]$
$\dot{m}_{eg}$	Exhaust Gases Airflow rate	67.82	$[\frac{kg}{s}]$
γ_{eg}	Exhaust gases' ratio of specific heat	1.3073	$[-]$
$c_{p,eg}$	Exhaust gases' pressure constant specific heat	1.216	$\left[\frac{kJ}{kg \cdot K}\right]$
$A_{out,hg}$	Hot-Nozzle exit section	0.51	$[m^2]$
$u_{out,hg}$	Hot-Nozzle exit velocity	554.20	$[\frac{m}{s}]$
$A_{out,cg}$	Cold-Nozzle exit section	6.14	$[m^2]$
$u_{out,cg}$	Cold-Nozzle exit velocity	303.38	$[\frac{m}{s}]$

Table 4: Resulting performance data

Table 4 summarizes the resulting performance data, providing detailed values for various parameters such as total thrust, hot and cold nozzle thrusts, $TSFC$, thermal and propulsive efficiency, airflow rates, and nozzle exit velocities. The data underscore the GE9X's enhanced performance metrics. For instance, the total thrust (F_{tot}) of $95,423N$ and a $TSFC$ of $0.0686kg/Nh$ reflect the engine's superior thrust capability and fuel efficiency. Additionally, the efficiencies ($\eta_{th} = 0.4827$, $\eta_{prop} = 0.6357$) and airflow rates demonstrate the advanced aerodynamic and thermodynamic design improvements of the GE9X over its predecessor, the GE90. This detailed dataset provides a quantitative foundation for assessing the GE9X's enhanced performance attributes.

Note: Reported thrust values and other performance data are referred to cruise altitude, which represents the primary operational condition for the engine. All calculations made are exposed in the following appendix through MATLAB code.

References

[1] *TYPE CERTIFICATE DATA SHEET.* https://drs.faa.gov/browse/excelExternalWindow/DRSDOCID114483736420230203181002.0001?modalOpened=true.

[2] Erik Dick. *Fundamentals of Turbomachines.* Springer, 2015.

[3] *GE90 vs GE9X: Which Boeing 777 Engine Type Is Most Powerful?* https://simpleflying.com/general-electric-boeing-777-engine-power-comparison.

[4] Michael Foust et al. *Development of the GE Aviation Low Emissions TAPS Combustor for Next Generation Aircraft Engines.* https://arc.aiaa.org/doi/abs/10.2514/6.2012-936.

[5] *Turbine Cooling Group.* https://oti.eng.ox.ac.uk/research/research-groups/turbine-cooling-group/.

[6] Saeed Farokhi. *Aircraft Propulsion.* Wiley, 2014.

[7] *GE9X Breaks GUINNESS WORLD RECORDS™ Title for Thrust.* https://www.geaerospace.com/news/press-releases/commercial-engines/ge9x-breaks-guinness-world-recordstm-title-thrust.

I Boeing 777X

The Boeing 777X represents a significant leap forward in commercial aviation, building on the proven success of the Boeing 777 series. This next-generation aircraft integrates cutting-edge aerodynamics and advanced materials, delivering enhanced performance and efficiency. The 777X comes in two variants, the 777-8 and 777-9, both offering increased capacity and range to meet evolving market demands. Key design innovations include a longer fuselage to accommodate more passengers and cargo, as well as revolutionary folding wingtips for superior aerodynamics and compatibility with existing airport infrastructure. Additionally, the Boeing 777X is powered by the state-of-the-art General Electric GE9X engines, setting new standards in fuel efficiency and environmental performance [cfr figure 20 and table 5].

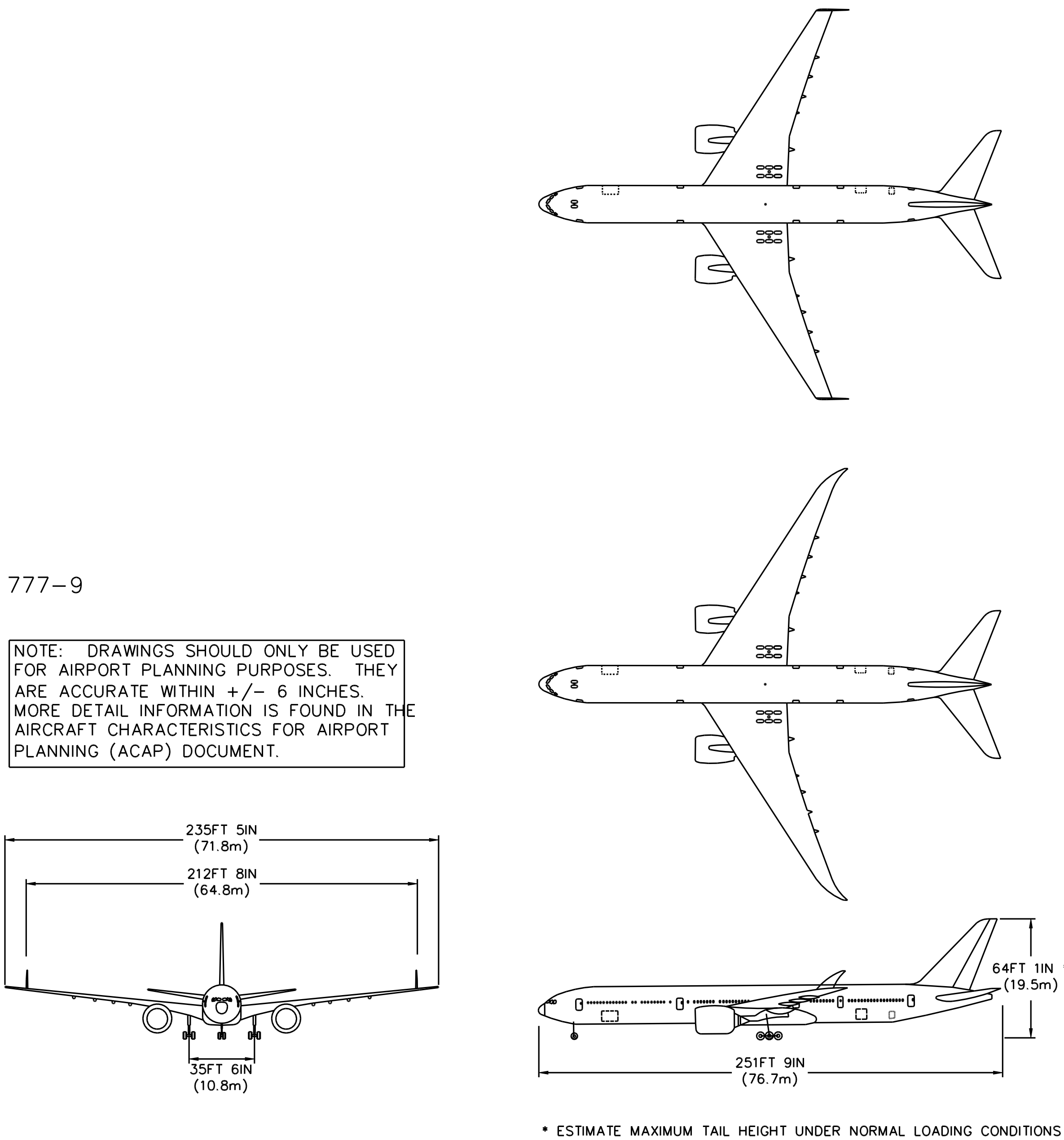

Figure 20: Blueprints for 777-9 model

Model	777-8	777-9	777-8F
Cockpit crew	Two		
Main deck 2-class seats	395	414-426 (42J + 372-384Y)	31 pallets (96" x 125")
Main deck 3-class seats	-	349-357 (8F + 49J + 292-300Y)	-
Lower deck ULD	40 LD3	48 LD-3: 26 fwd + 22 aft	13 pallets (96" x 125")
Cargo capacity	-	8,131 cu ft (230.2 m^3)	27,056 cu ft (766.1 m^3)
Length	232 ft 6 in (70.87 m)	251 ft 9 in (76.73 m)	232 ft 6 in (70.87 m)
Wingspan	235 ft 5 in (71.75 m), 212 ft 9 in (64.85 m) folded		
Wing	5,562 sq ft (516.7 m^2), 9.96 aspect ratio		
Height	63 ft 11 in (19.48 m)	64 ft 7 in (19.68 m)	64 ft (19.51 m)
Width	20 ft 4 in (6.20 m) fuselage, 19 ft 7 in (5.96 m) interior		
MTOW	775,000 lb (351.5 t)		805,000 lb (365.1 t)
Max. Payload	-	162,000 lb (73.5 t)	247,500 lb (112.3 t)
OEW	-	400,000 lb (180 t)	-
Fuel capacity	52,136 US gal (197,360 L; 43,412 imp gal)		
Ceiling*	43,100 ft (13,100 m)		
Speed*	Max. Mach 0.90 (516 kn; 956 km/h; 594 mph), Cruise Mach 0.85 (487 kn; 903 km/h; 561 mph)		
Engine ($\times$2)	General Electric GE9X-105B1A		-
Thrust ($\times$2)	110,000 lbf (489 kN)		-
Range	8,745 nmi (16,190 km; 10,050 mi)	7,285 nmi (13,500 km; 8,383 mi)	4,410 nmi (8,170 km; 5,070 mi)
ICAO designation	B778	B779	-

Table 5: 777X variants specs (* data are either guesstimates or refer to 777-300ER as are yet to be disclosed)

II Matlab® code

II.1 main.m

```
GE9X = data;
GE9X = inlet(GE9X);
GE9X = fan(GE9X);
GE9X = mach_fan(GE9X);
GE9X = lpc(GE9X);
GE9X = hpc(GE9X);
GE9X = CC(GE9X);
GE9X = hpt(GE9X);
GE9X = lpt(GE9X);
GE9X = nozzle_cold(GE9X);
GE9X = nozzle_hot(GE9X);
GE9X = cycle(GE9X);
GE9X = performance(GE9X);
fig1 = figure; % plotting exhaust speed Hot nozzle
plot(GE9X.amb.cruise.z/0.3048,GE9X.NOZZLE.HOT.u_out,'LineWidth',2)
hold on
grid on
box on
xlabel('Altitude $[ft]$','Interpreter','latex','FontSize',14)
ylabel('Speed $[m/s]$','Interpreter','latex','FontSize',14)
fig2 = figure; % plotting exhaust speed Cold nozzle
plot(GE9X.amb.cruise.z/0.3048,GE9X.NOZZLE.COLD.u_out,'LineWidth',2)
hold on
grid on
box on
xlabel('Altitude $[ft]$','Interpreter','latex','FontSize',14)
ylabel('Speed $[m/s]$','Interpreter','latex','FontSize',14)
T = [120, 140, 160, 180, 200, 220, 240, 260, 300, 320, 360, 400, 420, 460, 500,
    520, 560, 600, 650, 700, 750, 800, 850, 900, 950, 1000, 1050, 1100, 1150, 1200,
     1300, 1400, 1500, 1600];
cp = [1.02, 1.016, 1.013, 1.01, 1.008, 1.007, 1.006, 1.005, 1.006, 1.007, 1.01,
    1.015, 1.017, 1.024, 1.031, 1.035, 1.043, 1.051, 1.063, 1.075, 1.086, 1.098,
    1.109, 1.12, 1.131, 1.141, 1.15, 1.159, 1.167, 1.175, 1.188, 1.2, 1.21, 1.219];
[poli,s] = polyfit(T, cp, 4);
p = polyval(poli, T);
fig3 = figure; % Plotting Cp air vs T
plot(T,p,'LineWidth',1);
xlabel('Temperature $[K]$','Interpreter','latex','FontSize',14)
ylabel('Specific Heat $[kJ/kgK]$','Interpreter','latex','FontSize',14)
hold on
scatter(T, cp, 5,"red","LineWidth",1);
legend('approximate polynomial function', 'experimental points','Interpreter','
    latex','FontSize',14, Location='northwest')
```

II.2 data.m

```
function [GE9X] = data
GE9X.amb.std.P = 101325; % Pa
GE9X.amb.std.T = 288.15; % K
GE9X.amb.cruise.z = (5000:1000:39000)*0.3048; % m 777X cruise at FL390
GE9X.speed.cruise.M = 0.85; %extimated cruise mach form data sheet
GE9X.air.R = 8314/29; % J/kgK
GE9X.INLET.Diameter = 3.4; %m
```

```
GE9X.FAN.BPR = 9.9;
GE9X.FAN.num_blades = 16; %fan blade count
GE9X.FAN.core_speed = 2510*pi/30; % rad/s fan core speed
GE9X.FAN.pitch_line_radius = 1.65; % m fan pitch line radius
GE9X.COMPRESSOR.pitch_line_radius = 0.85; % m compressor pitch line radius
GE9X.COMPRESSOR.lpc_core_speed = 5500*pi/30;
GE9X.COMPRESSOR.hpc_core_speed = 11119*pi/30;
GE9X.COMPRESSOR.beta_tot = 60; %pressure ratio between HPC out and ambient
GE9X.COMPRESSOR.beta_comp = 27; %compressors pressure ratio
GE9X.COMPRESSOR.lpc_n_stages = 3; % LPC stages
GE9X.COMPRESSOR.hpc_n_stages = 11; % HPC stages
GE9X.CC.Q = 42.798e6; % J/kg Heat power
GE9X.CC.T_max = 1588.70556; % K 2400 F
GE9X.eg.gamma = 1.3073; % exhaust gases cp/cv
GE9X.eg.c_p = 1216; % J/kgK % exhaust gases cp
GE9X.TURBINE.T_out = 1365.15; % K maximum allowed temp at turbine exit
GE9X.INLET.pi_d = .980; % inlet pressure efficiency
GE9X.FAN.eta_f = .93; % Fan polytropic efficiency
GE9X.FAN.eta_f_m = .99; % Fan mechanic efficiency
GE9X.COMPRESSOR.LPC.eta_lpc = .91; % LPC polytropic efficiency
GE9X.COMPRESSOR.LPC.eta_lpc_m = .99; % LPC mechanic efficiency
GE9X.COMPRESSOR.HPC.eta_hpc = .91; % HPC polytropic efficiency
GE9X.COMPRESSOR.HPC.eta_hpc_m = .99; % HPC mechanic efficiency
GE9X.CC.eta_cc = .99; % CC polytropic effiency
GE9X.CC.pi_cc = .95; % CC pressure efficiency
GE9X.TURBINE.HPT.eta_hpt = .93; % HPT polytropic efficiency
GE9X.TURBINE.HPT.eta_hpt_m = .99; % HPT mechanic efficiency
GE9X.TURBINE.LPT.eta_lpt = .93; % LPT polytropic efficiency
GE9X.TURBINE.LPT.eta_lpt_m = .99; % LPT mechanic efficiency
GE9X.NOZZLE.HOT.eta_hn = .95; % Hot nozzle isentropic efficiency
GE9X.NOZZLE.COLD.eta_cn = .95; % Cold nozzle isentropic efficiency
end
```

II.3 inlet.m

```
function [GE9X] = inlet(GE9X)
R = GE9X.air.R;
lambda = -0.0065; % K/m
g = 9.81; % m/s^2
D = GE9X.INLET.Diameter; % HP inlet D == Fan D
pi_d = GE9X.INLET.pi_d;
M_in = GE9X.speed.cruise.M;
z_sgn = 11000; % end of troposphere
p_sgn = 22600; % static pressure at 11 km
T_sgn = 216.5; % static temperature above 11 km
for i = 1:length(GE9X.amb.cruise.z) % amb press and temp vs z
    if GE9X.amb.cruise.z(i) < z_sgn % troposphere
        T(i) = GE9X.amb.std.T + lambda*GE9X.amb.cruise.z(i);
        P(i) = GE9X.amb.std.P*(T(i)/GE9X.amb.std.T).^(-g/(R*lambda));
    else % stratosphere
        T(i) = T_sgn;
        P(i) = p_sgn*exp(-g*(GE9X.amb.cruise.z(i)-z_sgn)/(R*T_sgn));
    end
end
GE9X.amb.cruise.T = T;
GE9X.amb.cruise.P = P;
[~,gamma] = cp_air_T(T); % using cp model
T0_in = T.*(1 + ((gamma-1)./2).*M_in^2); % total T
```

```
GE9X.amb.cruise.T0 = T0_in;
P0_in = P.*(1 + ((gamma-1)./2).*M_in^2).^(gamma./(gamma-1)); % total P
GE9X.amb.cruise.P0 = P0_in;
GE9X.INLET.T0_in = T0_in;
GE9X.INLET.P0_in = P0_in;
rho = P./(T*R); % kg/m^3 air density at inlet
u = M_in * sqrt(gamma.*R.*T); % m/s inlet air speed
GE9X.speed.cruise.u = u;
m_a = rho.*u*(pi/4)*(D^2); % kg/s total air flow at inlet
GE9X.INLET.m_a = m_a;
GE9X.INLET.T0_out = T0_in;
GE9X.INLET.P0_out = pi_d * P0_in;
GE9X.INLET.beta_d = GE9X.INLET.P0_out./ P; % inlet p ratio
end
```

II.4 fan.m

```
function [GE9X] = fan(GE9X)
eta_f = GE9X.FAN.eta_f;
eta_f_m = GE9X.FAN.eta_f_m;
beta_f = GE9X.COMPRESSOR.beta_tot./GE9X.COMPRESSOR.beta_comp./GE9X.INLET.beta_d; %
    fan p ratio
GE9X.FAN.beta_f = beta_f;
T0_in = GE9X.INLET.T0_out;
[c_p,gamma] = cp_air_T(T0_in); % using cp model
P0_in = GE9X.INLET.P0_out;
GE9X.FAN.T0_in = T0_in;
GE9X.FAN.P0_in = P0_in;
m_a = GE9X.INLET.m_a; % kg/s fan air flow
GE9X.FAN.m_a = m_a;
T0_out_s = T0_in.*( beta_f.^((gamma-1)./gamma) ); %polytropic T0
GE9X.FAN.T0_out_s = T0_out_s;
T0_out = T0_in + (T0_out_s - T0_in)/eta_f; % real T0
GE9X.FAN.T0_out = T0_out;
P0_out = beta_f.*P0_in;
GE9X.FAN.P0_out = P0_out;
PWR_f = (1/eta_f_m).*c_p.*m_a.*(T0_out - T0_in); % fan required power
GE9X.FAN.PWR_f = PWR_f;
end
```

II.5 mach_fan.m

```
function [GE9X] = mach_fan(GE9X)
m = GE9X.FAN.m_a(end);
A = pi*(1.7)^2;
P = GE9X.FAN.P0_in(end);
T = GE9X.FAN.T0_in(end);
R = GE9X.air.R;
[cp,g] = cp_air_T(T);
F = @(M)((m*R)/(A*P))*(1+.5*(g-1)*M^2)^(1/(2*(g-1)))-M*sqrt((g-1)*cp/T);
dF = derivata(F,1); % function not included, calculates the derivative
phi = @(M)((((m*R)/(A*P))*(1+.5*(g-1)*M^2)^(1/(2*(g-1)))))/sqrt((g-1)*cp/T);
[M_b,~] = bisez(0,2,1e-5,F); % use bisection, f not included in appendix
[M_n,~] = newton(1,1000,1e-5,F,dF); % use newton, f not included
[M_p,~] = ptofis(1, phi, 1000, 1e-5); % use fixed point, f not included
```

```
GE9X.FAN.Mach_in = (M_p(end) + M_b(end) + M_n(end))/3; % in absence of above
    functions to run mach_fan impose directly in Velocity_triangles_fan.m script :
    GE9X.FAN.Mach_in = 0.6380, which is the result given from this function
```

II.6 lpc.m

```
function [GE9X] = lpc(GE9X)
eta_lpc = GE9X.COMPRESSOR.LPC.eta_lpc;
eta_lpc_m = GE9X.COMPRESSOR.LPC.eta_lpc_m;
beta_comp = GE9X.COMPRESSOR.beta_comp;
lpc_n_stages = GE9X.COMPRESSOR.lpc_n_stages;
hpc_n_stages = GE9X.COMPRESSOR.hpc_n_stages;
beta_lpc = beta_comp^(lpc_n_stages/(lpc_n_stages + hpc_n_stages));
GE9X.COMPRESSOR.LPC.beta_lpc = beta_lpc; % P ratio propto stage num
T0_in = GE9X.FAN.T0_out;
[c_p,gamma] = cp_air_T(T0_in); % using cp model
P0_in = GE9X.FAN.P0_out;
GE9X.COMPRESSOR.LPC.T0_in = T0_in;
GE9X.COMPRESSOR.LPC.P0_in = P0_in;
m_a = GE9X.INLET.m_a;
BPR = GE9X.FAN.BPR;
m_a_h = m_a/(1 + BPR); % hot air flow
GE9X.COMPRESSOR.LPC.m_a_h = m_a_h;
T0_out_s = T0_in.*( beta_lpc.^((gamma-1)./gamma) ); % iso T0
GE9X.COMPRESSOR.LPC.T0_out_s = T0_out_s;
T0_out = T0_in + (T0_out_s - T0_in)/eta_lpc; % real T0
GE9X.COMPRESSOR.LPC.T0_out = T0_out;
P0_out = beta_lpc*P0_in;
GE9X.COMPRESSOR.LPC.P0_out = P0_out;
PWR_lpc = (1/eta_lpc_m)*c_p.*m_a_h.*(T0_out - T0_in);
GE9X.COMPRESSOR.LPC.PWR_lpc = PWR_lpc; % LPC req power
end
```

II.7 hpc.m

```
function [GE9X] = hpc(GE9X)
eta_hpc = GE9X.COMPRESSOR.HPC.eta_hpc;
eta_hpc_m = GE9X.COMPRESSOR.HPC.eta_hpc_m;
beta_comp = GE9X.COMPRESSOR.beta_comp;
lpc_n_stages = GE9X.COMPRESSOR.lpc_n_stages;
hpc_n_stages = GE9X.COMPRESSOR.hpc_n_stages;
beta_hpc = beta_comp^(hpc_n_stages/(lpc_n_stages + hpc_n_stages));
GE9X.COMPRESSOR.HPC.beta_hpc = beta_hpc; % same as in LPC
T0_in = GE9X.COMPRESSOR.LPC.T0_out;
[c_p,gamma] = cp_air_T(T0_in); % using cp model
P0_in = GE9X.COMPRESSOR.LPC.P0_out;
GE9X.COMPRESSOR.HPC.T0_in = T0_in;
GE9X.COMPRESSOR.HPC.P0_in = P0_in;
m_a_h = GE9X.COMPRESSOR.LPC.m_a_h;
GE9X.COMPRESSOR.HPC.m_a_h = m_a_h;
T0_out_s = T0_in.*(beta_hpc.^((gamma-1)./gamma)); % iso T0
GE9X.COMPRESSOR.HPC.T0_out_s = T0_out_s; % real T0
T0_out = T0_in + (T0_out_s - T0_in)/eta_hpc;
GE9X.COMPRESSOR.HPC.T0_out = T0_out;
P0_out = beta_hpc*P0_in;
GE9X.COMPRESSOR.HPC.P0_out = P0_out;
PWR_hpc = (1/eta_hpc_m)*c_p.*m_a_h.*(T0_out - T0_in); % HPC req power
```

```
GE9X.COMPRESSOR.HPC.PWR_hpc = PWR_hpc;
end
```

II.8 CC.m

```
function [GE9X]=CC(GE9X)
eta_cc = GE9X.CC.eta_cc;
Pi_cc = GE9X.CC.pi_cc;
T0_in = GE9X.COMPRESSOR.HPC.T0_out;
P0_in = GE9X.COMPRESSOR.HPC.P0_out;
T_max = GE9X.CC.T_max; % K HP maximum temperature in cc always reached
Ma_h = GE9X.COMPRESSOR.HPC.m_a_h;
[Cp_a,~] = cp_air_T(T0_in);
Cp_gc = GE9X.eg.c_p;
Q = GE9X.CC.Q;
f = (-Cp_a.*T0_in+Cp_gc*T_max)./(eta_cc*Q-Cp_gc*T_max); % fuel/air ratio
P0_out = Pi_cc*P0_in;
Mf = f.*Ma_h; % kg/s Fuel flow rate
GE9X.CC.f = f;
GE9X.CC.T0_in = T0_in;
GE9X.CC.P0_in = P0_in;
GE9X.CC.T0_out = T_max;
GE9X.CC.P0_out = P0_out;
GE9X.CC.m_f = Mf;
end
```

II.9 hpt.m

```
function [GE9X] = hpt(GE9X)
gamma_eg = GE9X.eg.gamma;
c_p_eg = GE9X.eg.c_p;
PWR_hpc = GE9X.COMPRESSOR.HPC.PWR_hpc;
eta_hpt = GE9X.TURBINE.HPT.eta_hpt;
eta_hpt_m = GE9X.TURBINE.HPT.eta_hpt_m;
T0_in = GE9X.CC.T0_out;
P0_in = GE9X.CC.P0_out;
GE9X.TURBINE.HPT.T0_in = T0_in;
GE9X.TURBINE.HPT.P0_in = P0_in;
m_a_h = GE9X.COMPRESSOR.LPC.m_a_h;
f = GE9X.CC.f;
m_eg = m_a_h.*f; % kg/s exhaust gases flow
GE9X.TURBINE.HPT.m_a = m_a_h + m_eg;
T0_out = (m_a_h.*(1+f)*c_p_eg*eta_hpt_m.*T0_in-PWR_hpc)./(m_a_h.*(1+f)*c_p_eg*
    eta_hpt_m);
GE9X.TURBINE.HPT.T0_out = T0_out;
T0_out_s = T0_in+(T0_out-T0_in)/eta_hpt;
GE9X.TURBINE.HPT.T0_out_s = T0_out_s;
P0_out = P0_in.*(T0_in./T0_out_s).^(gamma_eg/(1-gamma_eg));
GE9X.TURBINE.HPT.P0_out = P0_out;
GE9X.TURBINE.HPT.beta_hpt = 1./(P0_out./P0_in); % expansion ratio
end
```

II.10 lpt.m

```
function [GE9X] = lpt(GE9X)
gamma_eg = GE9X.eg.gamma;
c_p_eg = GE9X.eg.c_p;
PWR_f = GE9X.FAN.PWR_f;
PWR_lpc = GE9X.COMPRESSOR.LPC.PWR_lpc;
eta_lpt = GE9X.TURBINE.LPT.eta_lpt;
eta_lpt_m = GE9X.TURBINE.LPT.eta_lpt_m;
T0_in = GE9X.TURBINE.HPT.T0_out;
P0_in = GE9X.TURBINE.HPT.P0_out;
GE9X.TURBINE.LPT.T0_in = T0_in;
GE9X.TURBINE.LPT.P0_in = P0_in;
T_max = GE9X.TURBINE.T_out; % K Indicated Turbine EG Temp CONSTRAINT
m_a_h = GE9X.COMPRESSOR.LPC.m_a_h;
f = GE9X.CC.f;
m_eg = m_a_h.*f;
GE9X.TURBINE.LPT.m_a = m_a_h + m_eg;
T0_out = (m_a_h.*(1+f)*c_p_eg*eta_lpt_m.*T0_in-(PWR_f+PWR_lpc))./(m_a_h.*(1+f)*
    c_p_eg*eta_lpt_m);
GE9X.TURBINE.LPT.T0_out = T0_out;
T0_out_s = T0_in+(T0_out-T0_in)/eta_lpt;
GE9X.TURBINE.LPT.T0_out_s = T0_out_s;
P0_out = P0_in.*(T0_in./T0_out_s).^(gamma_eg/(1-gamma_eg));
GE9X.TURBINE.LPT.P0_out = P0_out;
if any(T0_out > T_max) % max T check
    warning('Indicated Turbine Exhaust Gas Temperature exceeded')
end
GE9X.TURBINE.LPT.beta_lpt = 1./(P0_out./P0_in);
GE9X.TURBINE.beta_comp = GE9X.TURBINE.HPT.beta_hpt.*GE9X.TURBINE.LPT.beta_lpt; %
    expansion ratio
end
```

II.11 nozzle_cold.m

```
function [GE9X] = nozzle_cold(GE9X)
eta = GE9X.NOZZLE.COLD.eta_cn;
T0_in = GE9X.FAN.T0_out;
[cp,g] = cp_air_T(T0_in);
R    = 8314/29;
BPR = GE9X.FAN.BPR;
m_a_c = GE9X.INLET.m_a * BPR / (1 + BPR); % kg/s cold air flow
GE9X.NOZZLE.COLD.m_a_c = m_a_c;
P_amb = GE9X.amb.cruise.P;
P0_in = GE9X.FAN.P0_out;
P_out = zeros(1,length(P0_in));
T_out = zeros(1,length(P0_in));
T_out_s = zeros(1,length(P0_in));
P_crit = ( 2./(g+1) ) .^ ( g./(g-1) ) .* P0_in;
for ii = 1 : length(P0_in)
    if P_amb(ii)  > P_crit(ii) % no lock flow
        z = GE9X.amb.cruise.z(ii)/0.3048;
        fprintf('no lock flow at FL %f \n',z)
        P_out(ii) = P_amb(1); % note that cold nozzle always have lock flow
        T_out_s(ii) = T0_in(ii) * (P_out(ii)/P0_in(ii))^((g(ii)-1)/g(ii));
        T_out(ii)   = T0_in(ii) - (T0_in(ii) - T_out_s(ii))*eta;
        u_out(ii) = sqrt(2*cp(ii)*(T0_in(ii) - T_out(ii)));
        a(ii) = sqrt(g*R*T_out(ii));
        Mach(ii) = u_out(ii)/a(ii);
        rho_out(ii) = P_out(ii)/(R*T_out(ii));
```

```
        A_out(ii) = m_a_c(ii)/(rho_out(ii) * u_out(ii));
    else % lock flow
        P_out(ii) = P_crit(ii); % max P reached is P crit
        Mach(ii) = 1;
        T_out(ii) = 2*cp(ii)*T0_in(ii)/(R*g(ii)+2*cp(ii));
        T_out_s(ii) = T0_in(ii)+(T_out(ii)-T0_in(ii))/eta;
        rho_out(ii) = P_out(ii)/(R*T_out(ii));
        u_out(ii) = Mach(ii)*sqrt(g(ii)*R*T_out(ii));
        A_out(ii) = m_a_c(ii)/(rho_out(ii) * u_out(ii));
end
end
GE9X.NOZZLE.COLD.A_out     = A_out;
GE9X.NOZZLE.COLD.rho_out   = rho_out;
GE9X.NOZZLE.COLD.u_out     = u_out;
GE9X.NOZZLE.COLD.T0_in     = T0_in;
GE9X.NOZZLE.COLD.P0_in     = P0_in;
GE9X.NOZZLE.COLD.T_out     = T_out;
GE9X.NOZZLE.COLD.P_out     = P_out;
GE9X.NOZZLE.COLD.T_out_s   = T_out_s;
GE9X.NOZZLE.COLD.M_out     = Mach;
end
```

II.12 nozzle_hot.m

```
function [GE9X] = nozzle_hot(GE9X)
eta = GE9X.NOZZLE.HOT.eta_hn;
T0_in = GE9X.TURBINE.LPT.T0_out;
g = GE9X.eg.gamma;
cp  = GE9X.eg.c_p;
R = (g-1)/g*cp;
m_gc = GE9X.TURBINE.LPT.m_a;
GE9X.NOZZLE.HOT.m_gc = m_gc;
P_amb = GE9X.amb.cruise.P;
P0_in = GE9X.TURBINE.LPT.P0_out;
P_crit = ( 2/(g+1) ) ^ ( g/(g-1) ) * P0_in;
P_out = zeros(1,length(P0_in));
T_out = zeros(1,length(P0_in));
T_out_s = zeros(1,length(P0_in));
for ii = 1 : length(P0_in)
    if P_amb(ii)  > P_crit(ii)  % no lock flow
        z = GE9X.amb.cruise.z(ii)/0.3048;
        fprintf('no lock flow at FL %f \n',z)
        P_out(ii) = P_amb(1); % note that nozzle always have lock flow
        T_out_s(ii) = T0_in(ii) * (P_out(ii)/P0_in(ii))^((g-1)/g);
        T_out(ii)   = T0_in(ii) - (T0_in(ii) - T_out_s(ii))*eta;
        u_out(ii) = sqrt(2*cp*(T0_in(ii) - T_out(ii)));
        a(ii) = sqrt(g*R*T_out(ii));
        Mach(ii) = u_out(ii)/a(ii);
        rho_out(ii) = P_out(ii)/(R*T_out(ii));
        A_out(ii) = m_gc(ii)/(rho_out(ii) * u_out(ii));
    else  % max P reached is P crit
        P_out(ii) = P_crit(ii);
        Mach(ii) = 1;
        T_out(ii) = 2*cp*T0_in(ii)/(R*g+2*cp);
        T_out_s(ii) = T0_in(ii)+(T_out(ii)-T0_in(ii))/eta;
        rho_out(ii) = P_out(ii)/(R*T_out(ii));
        u_out(ii) = Mach(ii)*sqrt(g*R*T_out(ii));
        A_out(ii) = m_gc(ii)/(rho_out(ii) * u_out(ii));
```

```
end
end
GE9X.NOZZLE.HOT.A_out    = A_out;
GE9X.NOZZLE.HOT.rho_out  = rho_out;
GE9X.NOZZLE.HOT.u_out    = u_out;
GE9X.NOZZLE.HOT.T0_in    = T0_in;
GE9X.NOZZLE.HOT.P0_in    = P0_in;
GE9X.NOZZLE.HOT.T_out    = T_out;
GE9X.NOZZLE.HOT.P_out    = P_out;
GE9X.NOZZLE.HOT.T_out_s  = T_out_s;
GE9X.NOZZLE.HOT.M_out    = Mach;
end
```

II.13 performance.m

```
function [GE9X] = performance(GE9X)
g0    = 9.81;
u     = GE9X.speed.cruise.u;
ma    = GE9X.INLET.m_a;
ma_h  = GE9X.COMPRESSOR.LPC.m_a_h;
f     = GE9X.CC.f;
Q     = GE9X.CC.Q;
u_hot = GE9X.NOZZLE.HOT.u_out;
A_hot = GE9X.NOZZLE.HOT.A_out;
P_hot = GE9X.NOZZLE.HOT.P_out;
P_amb = GE9X.amb.cruise.P;
BPR   = GE9X.FAN.BPR;
u_cold= GE9X.NOZZLE.COLD.u_out;
A_cold= GE9X.NOZZLE.COLD.A_out;
P_cold= GE9X.NOZZLE.COLD.P_out;
F_hot = ma_h.*(1+f).*u_hot + A_hot.*(P_hot-P_amb); % Only hot nozz thrust
F_cold= ma_h.*BPR.*u_cold + A_cold.*(P_cold-P_amb); % Only cold nozz thrust
F_tot = F_hot + F_cold - ma.*u; % total thrust with ram drag counted
GE9X.performance.F_tot_calculated  = F_tot;
GE9X.performance.F_hot_calculated  = F_hot;
GE9X.performance.F_cold_calculated = F_cold;
TSFC_hr = 3600*f.*ma_h./F_tot; % Kg/Nhr thrust specific fuel consumption
GE9X.performance.I_sp = I_sp;
GE9X.performance.TSFC_hr = TSFC_hr;
u_hot_eff = F_hot./(ma_h.*(1+f)); % effective hot air speed
if BPR == 0
    u_cold_eff = 0; % No cold air running though nozzle
else
    u_cold_eff = F_cold./(ma_h.*BPR); % effective cold air speed
end
eta_th = (ma_h.*BPR.*u_cold_eff.^2 + ma_h.*(1+f).*u_hot_eff.^2 - (ma_h.*BPR+ma_h)
    .*u.^2)./(2*ma_h.*f*Q);
eta_prop = 2*F_tot.*u./(ma_h.*BPR.*u_cold_eff.^2 + ma_h.*(1+f).*u_hot_eff.^2 - (
    ma_h.*BPR+ma_h).*u.^2);
eta_gl = eta_th.*eta_prop;
GE9X.performance.eta_th = eta_th; % thermal efficiency
GE9X.performance.eta_prop = eta_prop; % Propulsive efficiency
GE9X.performance.eta_gl = eta_gl; % global efficiency
end
```

II.14 cp_air_T.m

```
function [cp,gamma] = cp_air_T(temp)
R = 8314/29;
T = [120, 140, 160, 180, 200, 220, 240, 260, 300, 320, 360, 400, 420, 460, 500,
    520, 560, 600, 650, 700, 750, 800, 850, 900, 950, 1000, 1050, 1100, 1150, 1200,
     1300, 1400, 1500, 1600];
cp_val = [1.02, 1.016, 1.013, 1.01, 1.008, 1.007, 1.006, 1.005, 1.006, 1.007,
    1.01, 1.015, 1.017, 1.024, 1.031, 1.035, 1.043, 1.051, 1.063, 1.075, 1.086,
    1.098, 1.109, 1.12, 1.131, 1.141, 1.15, 1.159, 1.167, 1.175, 1.188, 1.2, 1.21,
    1.219];
[poli,~] = polyfit(T, cp_val, 4); % fitting curve for given data
cp = polyval(poli, temp)*1e3; % cp value at requested Temp
cv = cp - R;
gamma = (cp./cv); % gamma value at requested temp
end
```

II.15 Velocity_triangles_fan.m

```
T01 = GE9X.FAN.T0_in(end);
P01 = GE9X.FAN.P0_in(end);
T02 = GE9X.FAN.T0_out(end);
P02 = GE9X.FAN.P0_out(end);
M1 = GE9X.FAN.Mach_in;
R = GE9X.air.R;
m_a = GE9X.INLET.m_a(end);
N = GE9X.FAN.num_blades;
omega_fan = GE9X.FAN.core_speed;
hub_to_tip_ratio = 0.3; % typical hub value
[cp1,g1] = cp_air_T(T01); % using cp model
T1 = T01/(1 + 0.5*(g1-1)*M1^2);
a1 = sqrt(g1*R*T1);
u1 = M1*a1;
P1 = P01*(T1/T01)^(g1/(g1-1));
rho1 = P1/(R*T1);
r_tip = sqrt(m_a/(pi*rho1*u1*(1 - hub_to_tip_ratio^2)));
r_hub = hub_to_tip_ratio*r_tip;
r_m = 0.5*(r_hub + r_tip);
A1 = pi*(r_tip^2 - r_hub^2);
h1 = r_tip - r_hub;
U_tip = omega_fan*r_tip;
w1 = sqrt(u1^2 + U_tip^2);
M1_rel = w1/a1; % super sonic check
[cp2,g2] = cp_air_T(T02);
T2 = T02 - (u1^2)/(2*cp2);
P2 = P02*(T2/T02)^(g2/(g2-1));
rho2 = P2/(R*T2);
A2 = m_a/(rho2*u1);
cp_m = 0.5*(cp1+cp2); % total energy exchange
dh0_fan = cp_m*(T02-T01);
h2 = A2/(2*pi*r_m); % extimating blade height at exit with rm = const
r_tip2 = r_m + 0.5*h2; % values at exit
r_hub2 = r_m - 0.5*h2;
hub_to_tip_ratio2 = r_hub2/r_tip2;
Utip2 = omega_fan*r_tip2;
Um = omega_fan*r_m;
alpha1 = 0; % per hp % reaction degree
dh0_st = dh0_fan; % fan has one stage
u2m_t = dh0_st/Um; % tangential component
```

```
u2m = sqrt(u1^2 + u2m_t^2); % absolute speed
w2m_t = Um - u2m_t; % tangential component
w2m = sqrt(w2m_t^2 + u1^2); % relative speed at exit
w1m = sqrt(u1^2 + Um^2); % relative speed at inlet at mean radius
w1m_t = Um; % tangential component
deHaller_fan = w2m/w1m; % De Haller is validated
deHaller_fan2 = u1/u2m; % De Haller is validated
tg_beta1 = w1m_t/u1; % angles
beta1 = atan(tg_beta1);
beta1_dg = beta1*180/pi;
tg_beta2 = w2m_t/u1;
beta2 = atan(tg_beta2);
beta2_dg = beta2*180/pi;
tg_alpha2 = u2m_t/u1;
alpha2 = atan(tg_alpha2);
alpha2_dg = alpha2*180/pi;
R = (u1/(2*Um))*(tg_beta1 + tg_beta2); % fan reaction degree
phi1 = u1/Um;
psi = 1 - phi1*(tg_beta2);
Ro = 0.5 + 0.5*phi1*tg_beta2; % verifying if R is the same
Re = 1200000; % typical Reynolds
nu1 = 8.53*1e-5; % viscosity
cm = Re*nu1/w1m;
sigma = N*cm/(2*pi*r_m); % sigma<0.7->acceptable for isolated foil theory
Dr = 1 - w2m/w1m + abs(w2m_t - w1m_t)/(2*sigma*w1m);
Ds = 1 - u1/u2m + abs( -u2m_t)./(2*sigma*u2m);
```

II.16 Velocity_triangles_lpc.m

```
T01 = GE9X.COMPRESSOR.LPC.T0_in(end);
P01 = GE9X.COMPRESSOR.LPC.P0_in(end);
T02 = GE9X.COMPRESSOR.LPC.T0_out(end);
P02 = GE9X.COMPRESSOR.LPC.P0_out(end);
R = GE9X.air.R;
m_a = GE9X.COMPRESSOR.LPC.m_a_h(end);
omega_lpc = GE9X.COMPRESSOR.lpc_core_speed;
hub_to_tip_ratio = 0.5; % typical hub to tip ratio
[cp1,g1] = cp_air_T(T01);
u1 = 193.3487049742902; % value from precedent analysis
T1 = T01 - (u1^2)/(2*cp1);
a1 = sqrt(g1*R*T1);
P1 = P01*(T1/T01)^(g1/(g1-1));
rho1 = P1/(R*T1);
r_tip = sqrt(m_a/(pi*rho1*u1*(1 - hub_to_tip_ratio^2)));
r_hub = hub_to_tip_ratio*r_tip;
r_m = 0.5*(r_hub + r_tip);
A1 = pi*(r_tip^2 - r_hub^2);
h1 = r_tip - r_hub;
U_tip = omega_lpc*r_tip;
w1 = sqrt(u1^2 + U_tip^2);
M1_rel = w1/a1; % transonic mach
[cpf2,gf2] = cp_air_T(T02);
cp_m = 0.5*(cp1+cpf2); % total energy exchange
dh0_lpc = cp_m*(T02-T01);
dh0_st = dh0_lpc/3; % lpc has 3 stages
dT_st = (T02-T01)/3;
T0f = T01 + dT_st;
P0f = P01*(T0f/T01)^(g1/(g1-1));
```

```
[cp2,g2] = cp_air_T(T0f);
T2 = T0f - (u1^2)/(2*cp2); % exit values
P2 = P0f*(T2/T0f)^(g2/(g2-1));
rho2 = P2/(R*T2);
A2 = m_a/(rho2*u1);
h2 = A2/(2*pi*r_m); % blade height with rm = const
r_tip2 = r_m + 0.5*h2; % at exit
r_hub2 = r_m - 0.5*h2;
hub_to_tip_ratio2 = r_hub2/r_tip2;
Utip2 = omega_lpc*r_tip2;
Um = omega_lpc*r_m;
alpha1 = 0; % hypothesis
u2m_t = dh0_st/Um; % tangential speed
u2m = sqrt(u1^2 + u2m_t^2); % absolute speed at mean radius
w2m_t = Um - u2m_t; % relative speed at mean radius
w2m = sqrt(w2m_t^2 + u1^2); % tangential relative speed
w1m = sqrt(u1^2 + Um^2);
w1m_t = Um; % tangential component
deHaller_lpc = w2m/w1m; % De haller is validated
deHaller_lpc2 = u1/u2m; % De haller is validated
tg_beta1 = w1m_t/u1; % angles
beta1 = atan(tg_beta1);
beta1_dg = beta1*180/pi;
tg_beta2 = w2m_t/u1;
beta2 = atan(tg_beta2);
beta2_dg = beta2*180/pi;
tg_alpha2 = u2m_t/u1;
alpha2 = atan(tg_alpha2);
alpha2_dg = alpha2*180/pi;
R = (u1/(2*Um))*(tg_beta1 + tg_beta2); % lpc reaction degree
phi1 = u1/Um;
psi = 1 - phi1*(tg_beta2);
Ro = 0.5 + 0.5*phi1*tg_beta2; % R is verified to be the same
Re = 410000; % guessed Re
nu1 = 8.53*1e-5;
cm = Re*nu1/w1m;
sigma_prov = 1.2;
N = round(sigma_prov*(2*pi*r_m)/cm); % N=28, using this value for stator stage
sigma_r = N*cm/(2*pi*r_m);
sigma_s = 1;
cs = sigma_s*(2*pi*r_m)/N;
Dr = 1 - w2m/w1m + abs(w2m_t - w1m_t)./(2*sigma_r*w1m);
Ds = 1 - u1/u2m + abs( -u2m_t)./(2*sigma_s*u2m);
```

II.17 Velocity_triangles_hpc.m

```
T01 = GE9X.COMPRESSOR.HPC.T0_in(end);
P01 = GE9X.COMPRESSOR.HPC.P0_in(end);
T02 = GE9X.COMPRESSOR.HPC.T0_out(end);
P02 = GE9X.COMPRESSOR.HPC.P0_out(end);
R = GE9X.air.R;
m_a = GE9X.COMPRESSOR.HPC.m_a_h(end);
omega_hpc = GE9X.COMPRESSOR.hpc_core_speed;
hub_to_tip_ratio = 0.5; % typical hub to tip ratio
[cp1,g1] = cp_air_T(T01);
u1 = 193.3487049742902;   % value from precedent analysis
T1 = T01 - (u1^2)/(2*cp1);
a1 = sqrt(g1*R*T1);
```

```
P1 = P01*(T1/T01)^(g1/(g1-1));
rho1 = P1/(R*T1);
r_tip = sqrt(m_a/(pi*rho1*u1*(1 - hub_to_tip_ratio^2)));
r_hub = hub_to_tip_ratio*r_tip;
r_m = 0.5*(r_hub + r_tip);
A1 = pi*(r_tip^2 - r_hub^2);
h1 = r_tip - r_hub;
U_tip = omega_hpc*r_tip;
w1 = sqrt(u1^2 + U_tip^2);
M1_rel = w1/a1; % transonic mach
[cpf2,gf2] = cp_air_T(T02);
cp_m = 0.5*(cp1+cpf2); % total energy exchange
dh0_hpc = cp_m*(T02-T01);
dh0_st = dh0_hpc/11; % hpc has 11 stages
dT_st = (T02-T01)/11;
T0f = T01 + dT_st;
P0f = P01*(T0f/T01)^(g1/(g1-1));
[cp2,g2] = cp_air_T(T0f);
T2 = T0f - (u1^2)/(2*cp2); % at exit
P2 = P0f*(T2/T0f)^(g2/(g2-1));
rho2 = P2/(R*T2);
A2 = m_a/(rho2*u1);
h2 = A2/(2*pi*r_m); % blade height with rm = const
r_tip2 = r_m + 0.5*h2; % at exit
r_hub2 = r_m - 0.5*h2;
hub_to_tip_ratio2 = r_hub2/r_tip2;
Utip2 = omega_hpc*r_tip2;
Um = omega_hpc*r_m;
alpha1 = 0;  % hypothesis
u2m_t = dh0_st/Um; % tangential component
u2m = sqrt(u1^2 + u2m_t^2); % absolute speed at mean radius
w2m_t = Um - u2m_t; % relative speed tangential component
w2m = sqrt(w2m_t^2 + u1^2); % relative speed at exit
w1m = sqrt(u1^2 + Um^2); % relative speed at entrance at mean radius
w1m_t = Um; % tangential componente
deHaller_hpc = w2m/w1m; % De Haller is validated
deHaller_hpc2 = u1/u2m; % De Haller is validated
tg_beta1 = w1m_t/u1; % angles
beta1 = atan(tg_beta1);
beta1_dg = beta1*180/pi;
tg_beta2 = w2m_t/u1;
beta2 = atan(tg_beta2);
beta2_dg = beta2*180/pi;
tg_alpha2 = u2m_t/u1;
alpha2 = atan(tg_alpha2);
alpha2_dg = alpha2*180/pi;
R = (u1/(2*Um))*(tg_beta1 + tg_beta2); % hpc Reaction degree
phi1 = u1/Um;
psi = 1 - phi1*(tg_beta2);
Ro = 0.5 + 0.5*phi1*tg_beta2; % R is verified to be the same
Re = 320000;
nu1 = 7.53*1e-5;
cm = Re*nu1/w1m;
sigma_prov = 1; % imposing sigma = 1
N = round(sigma_prov*(2*pi*r_m)/cm); % N=36, using this value for stator stage
sigma_r = N*cm/(2*pi*r_m);
sigma_s = 1.2;
cs = sigma_s*(2*pi*r_m)/N;
Dr = 1 - w2m/w1m + abs(w2m_t - w1m_t)./(2*sigma_r*w1);
```

```
Ds = 1 - u1/u2m + abs( -u2m_t)./(2*sigma_s*u2m);
```

II.18 CEA_analysis.m

```
OF = 20:1:70; % Data taken from NASA CEA software
T = [2044.69,1983.83,1925.70,1870.73,1818.97,1770.30,1724.54,1681.49,1640.95,...
1602.71,1566.59,1532.43,1500.08,1469.39,1440.25,1412.53,1386.14,1360.98,1336.97,...
1314.04,1292.10,1271.09,1250.96,1231.66,1213.12,1195.31,1178.19,1161.71,1145.84,...
1130.54,1115.78,1101.54,1087.79,1074.50,1061.65,1049.22,1037.19,1025.53,1014.23,...
1003.28,992.66,982.35,972.34,962.61,953.16,943.97,935.04,926.34,917.88,909.64,...
901.61]-273.15;
fig1 = figure;
plot(OF , T , LineWidth=1 , Color='k')
grid on
xlabel('O/F', 'Interpreter', 'latex', 'FontSize', 14)
ylabel('Temperature [$^\circ$C]', 'Interpreter', 'latex', 'FontSize', 14)
var1 = [];
for i = 2:length(T)
    var1 = [ var1 (T(i-1)-T(i))/T(i-1)];
end
var1 = var1*100;
var3 = [];
for i = 4:length(T)
    var3 = [ var3 ((T(i-3)+T(i-2)+T(i-1))/3-T(i))/((T(i-3)+T(i-2)+T(i-1))/3)];
end
var3 = var3*100;
var5 = [];
for i = 6:length(T)
    var5 = [ var5 ((T(i-4)+T(i-5)+T(i-3)+T(i-2)+T(i-1))/5-T(i))/((T(i-4)+T(i-5)+T(
        i-3)+T(i-2)+T(i-1))/5)];
end
var5 = var5*100;
fig2 = figure;
plot( OF(2:end) , var1 , LineWidth=1 , Color='k')
grid on
xlabel( 'O/F', 'Interpreter', 'latex', 'FontSize', 14)
ylabel('Gradient $\%$', 'Interpreter', 'latex', 'FontSize', 14)
hold on
plot( OF(4:end) , var3 , LineWidth=1 , Color='r')
plot( OF(6:end) , var5 , LineWidth=1 , Color='b')
legend('Deviation from previous OF value','Deviation from the average of the
    previous 3 values','Deviation from the average of the previous 5 values',...
'Interpreter', 'latex', 'FontSize', 12)
```

II.19 Nozzle_analysis.m

```
g0    = 9.81; % [m/s^2]
u     = GE9X.speed.cruise.u(end);
ma    = GE9X.INLET.m_a(end);
ma_h  = GE9X.COMPRESSOR.LPC.m_a_h(end);
ma_c  = GE9X.NOZZLE.COLD.m_a_c(end);
f     = GE9X.CC.f(end);
Q     = GE9X.CC.Q(end);
u_hot = GE9X.NOZZLE.HOT.u_out(end);
A_hot = GE9X.NOZZLE.HOT.A_out(end);
P_hot = GE9X.NOZZLE.HOT.P_out(end);
```

```
BPR    = GE9X.FAN.BPR;
u_cold= GE9X.NOZZLE.COLD.u_out(end);
A_cold= GE9X.NOZZLE.COLD.A_out(end);
P_cold= GE9X.NOZZLE.COLD.P_out(end);
cp_eg = GE9X.eg.c_p;
g_eg = GE9X.eg.gamma;
T0_in_hot = GE9X.NOZZLE.HOT.T0_in(end);
P0_in_hot = GE9X.NOZZLE.HOT.P0_in(end);
P_amb = GE9X.amb.cruise.P(end);
F_conv_hot = GE9X.performance.F_hot_calculated(end); % convergent nozzle Thrust
u_dl_hot = sqrt(2*cp_eg*T0_in_hot.*(1-(P_amb./P0_in_hot).^((g_eg-1)/g_eg)));
F_dl_hot = ma_h.*u_dl_hot; % Equivalent De laval
F_ratio_Hot = F_dl_hot./F_conv_hot;
NPR_hot =  P0_in_hot./P_amb; % Nozzle pressure ratio
T0_in_cold = GE9X.NOZZLE.COLD.T0_in(end);
P0_in_cold = GE9X.NOZZLE.COLD.P0_in(end);
[cp_air,g_air] = cp_air_T(T0_in_cold); % using cp model
F_conv_cold = GE9X.performance.F_cold_calculated(end); % cold nozzle
u_dl_cold = sqrt(2*cp_air.*T0_in_cold.*(1-(P_amb./P0_in_cold).^((g_air-1)./g_air))
    );
F_dl_cold = ma_c.*u_dl_cold; % equivalent de laval nozzle
F_ratio_cold = F_dl_cold./F_conv_cold;
NPR_cold = P0_in_cold./P_amb;
NPR = 2:0.1:20;
F = @(NPR,g) sqrt((1-NPR.^(-(g-1)/g))./((g-1)/(g+1))).*g./(g+(1-((g+1)/2).^(g/(g
    -1)).*NPR.^-1));
plot(NPR,F(NPR,g_eg),'red',NPR,F(NPR,g_air(end)),'blu',[2 10],[1.05 1.05],'black')
xlabel('Nozzle pressure ratio ($NPR$)','Interpreter','latex','FontSize',14)
ylabel('$F_{De Laval}/F_{convergent}$','Interpreter','latex','FontSize',14)
legend('\gamma_{eg} = 1.3073','\gamma_{air} = 1.3987','Location','best')
```

II.20 BPR_analysis.m

```
z_test = 39000*0.3048; % analysis at cruise level
GE9X = data;
GE9X.amb.cruise.z = z_test;
BPR_true = GE9X.FAN.BPR;
BPR_max = 14; % maximum BPR chosen
BPR_vect = 0:1:BPR_true;
i_true = length(BPR_vect) + 1;
BPR_vect = [BPR_vect, BPR_true, ceil(BPR_true):1:BPR_max];
GE9X_vect = [];
for BPR = BPR_vect % cycling to get renewed results for each BPR
    GE9X.FAN.BPR = BPR;
    GE9X = inlet(GE9X);
    GE9X = fan(GE9X);
    GE9X = lpc(GE9X);
    GE9X = hpc(GE9X);
    GE9X=CC(GE9X);
    GE9X = hpt(GE9X);
    GE9X = lpt(GE9X);
    GE9X = nozzle_cold(GE9X);
    GE9X = nozzle_hot(GE9X);
    GE9X = performance(GE9X);
    GE9X_vect = [GE9X_vect GE9X];
end
i_max = length(GE9X_vect);
m_a_vect = GE9X_vect(1).INLET.m_a * ones(1, i_max);
```

```
m_a_h_vect = m_a_vect ./ (1 + BPR_vect);
F_vect = [];
f_vect = [];
TSFC_vect = [];
eta_th_vect = [];
eta_prop_vect = [];
eta_gl_vect = [];
for i = 1 : i_max
    F_vect = [F_vect GE9X_vect(i).performance.F_tot_calculated];
    f_vect = [f_vect GE9X_vect(i).CC.f];
    TSFC_vect = [TSFC_vect GE9X_vect(i).performance.TSFC_hr];
    eta_th_vect = [eta_th_vect GE9X_vect(i).performance.eta_th];
    eta_prop_vect = [eta_prop_vect GE9X_vect(i).performance.eta_prop];
    eta_gl_vect = [eta_gl_vect GE9X_vect(i).performance.eta_gl];
end
m_f_vect = f_vect .* m_a_h_vect;
TS_air = F_vect ./ m_a_vect; % Thrust specific to fan air flow
TS_air_hot = F_vect ./ m_a_h_vect; % Thrust specific to hot air cyc
lineColor = [.5 .5 .5];
BPR_vects = BPR_vect(1:i_true-1);
BPR_vects = [BPR_vects, BPR_vect(i_true + 2 : end)];
TS_airs = TS_air(1:i_true-1);
TS_airs = [TS_airs, TS_air(i_true + 2 : end)];
TS_air_hots = TS_air_hot(1:i_true - 1);
TS_air_hots = [TS_air_hots, TS_air_hot(i_true + 2 : end)];
fig1 = figure;
plot(BPR_vect, TS_air, '-', BPR_vect, TS_air_hot, '-', 'LineWidth', 1.2);
hold on;
scatter(BPR_vects, TS_airs, 25, 'blue');
scatter(BPR_vects, TS_air_hots, 25, 'red');
scatter(BPR_vect(i_true), TS_air(i_true), 25, 'k', 'LineWidth', 1.2);
scatter(BPR_vect(i_true), TS_air_hot(i_true), 25, 'k', 'LineWidth', 1.2);
line([BPR_vect(i_true) BPR_vect(i_true)],[0 TS_air_hot(i_true)],...
     'Color', 'k', 'LineStyle', '--', 'Color', lineColor);
line([0 BPR_vect(i_true)],[TS_air_hot(i_true) TS_air_hot(i_true)],...
     'Color', 'k', 'LineStyle', '--', 'Color', lineColor);
line([0 BPR_vect(i_true)],[TS_air(i_true) TS_air(i_true)],...
     'Color', 'k', 'LineStyle', '--', 'Color', lineColor);
legend('F/$\dot{m}_{a}$', 'F/$\dot{m}_{a, hot}$', 'Interpreter',...
       'latex', fontsize=14, Location='northwest');
xlabel('BPR $[-]$','Interpreter', 'latex', fontsize=14);xticks([0 1 2 3 4 5 6 7 8
   9 9.9 11 12 13 14])
xticklabels({'0','1','2','3','4','5','6','7','8','9','9.9','11','12','13','14'})
ylabel('Specific thrust $[m/s]$','Interpreter', 'latex', fontsize=14);
TSFC_vects = TSFC_vect(1:i_true-1);
TSFC_vects = [TSFC_vects, TSFC_vect(i_true+2 : end)];
fig2 = figure;
plot(BPR_vect, TSFC_vect, '-', 'LineWidth', 1.2);
hold on;
scatter(BPR_vects, TSFC_vects, 25, 'blue');
scatter(BPR_vect(i_true), TSFC_vect(i_true), 25, 'k', 'LineWidth', 1.2);
line([BPR_vect(i_true) BPR_vect(i_true)],[0.05 TSFC_vect(i_true)],...
     'Color', 'k', 'LineStyle', '--', 'Color', lineColor);
line([0 BPR_vect(i_true)],[TSFC_vect(i_true) TSFC_vect(i_true)],...
     'Color', 'k', 'LineStyle', '--', 'Color', lineColor);
xlabel('BPR $[-]$','Interpreter', 'latex', fontsize=14);
xticks([0 1 2 3 4 5 6 7 8 9 9.9 11 12 13 14])
xticklabels({'0','1','2','3','4','5','6','7','8','9','9.9','11','12','13','14'})
ylabel('TSFC $[\frac{Kg}{N\cdot h}]$','Interpreter', 'latex', fontsize=14);
```

```
eta_prop_vects = eta_prop_vect(1:i_true-1);
eta_prop_vects = [eta_prop_vects, eta_prop_vect(i_true+2 : end)];
eta_th_vects = eta_th_vect(1:i_true-1);
eta_th_vects = [eta_th_vects, eta_th_vect(i_true+2 : end)];
eta_gl_vects = eta_gl_vect(1:i_true-1);
eta_gl_vects = [eta_gl_vects, eta_gl_vect(i_true+2 : end)];
blueColor = [0 0.4470 0.7410];
redColor = [0.8500 0.3250 0.0980];
yellowColor = [0.77 0.55 0];
fig3 = figure;
plot(BPR_vect, eta_prop_vect, '-', "Color", blueColor, 'LineWidth', 1.2)
hold on;
plot(BPR_vect, eta_th_vect, '-', "Color", redColor, 'LineWidth', 1.2);
plot(BPR_vect, eta_gl_vect, '-', "Color", yellowColor, 'LineWidth', 1.2);
scatter(BPR_vects, eta_prop_vects, 25, 'blue');
scatter(BPR_vects, eta_th_vects, 25, 'red');
scatter(BPR_vects, eta_gl_vects, 25, 'MarkerEdgeColor', yellowColor);
scatter(BPR_vect(i_true), eta_prop_vect(i_true), 25, 'k', 'LineWidth', 1.2);
scatter(BPR_vect(i_true), eta_th_vect(i_true), 25, 'k', 'LineWidth', 1.2);
scatter(BPR_vect(i_true), eta_gl_vect(i_true), 25, 'k', 'LineWidth', 1.2);
line([BPR_vect(i_true) BPR_vect(i_true)],[0 eta_prop_vect(i_true)],...
     'Color', 'k', 'LineStyle', '--', 'Color', lineColor);
line([0 BPR_vect(i_true)],[eta_th_vect(i_true) eta_th_vect(i_true)],...
     'Color', 'k', 'LineStyle', '--', 'Color', lineColor);
line([0 BPR_vect(i_true)],[eta_prop_vect(i_true) eta_prop_vect(i_true)],...
     'Color', 'k', 'LineStyle', '--', 'Color', lineColor);
line([0 BPR_vect(i_true)],[eta_gl_vect(i_true) eta_gl_vect(i_true)],...
     'Color', 'k', 'LineStyle', '--', 'Color', lineColor);
xlabel('BPR $[-]$','Interpreter', 'latex', fontsize=14);
xticks([0 1 2 3 4 5 6 7 8 9 9.9 11 12 13 14])
xticklabels({'0','1','2','3','4','5','6','7','8','9','9.9','11','12','13','14'})
ylabel('Efficiency $[-]$','Interpreter', 'latex', fontsize=14);
legend('\eta_p_r_o_p', '\eta_t_h', '\eta_g_l', fontsize=11,...
       Location='best');
```

II.21 Temp_Press_vs_Stage.m

```
T01 = GE9X.amb.cruise.T0(end);
T02 = GE9X.INLET.T0_out(end);
T03 = GE9X.FAN.T0_out(end);
T04 = GE9X.COMPRESSOR.LPC.T0_out(end);
T05 = GE9X.COMPRESSOR.HPC.T0_out(end);
T06 = GE9X.CC.T0_out(end);
T07 = GE9X.TURBINE.HPT.T0_out(end);
T08 = GE9X.TURBINE.LPT.T0_out(end);
T9 = GE9X.NOZZLE.HOT.T_out(end);
T10 = GE9X.NOZZLE.COLD.T_out(end);
fig1 = figure;
plot([1 2 3 4 5 6 7 8 9],[T01 T02 T03 T04 T05 T06 T07 T08 T9],"red",[3 9],[T03 T10
    ],"blu")
xlabel("Stage","Interpreter","latex","FontSize",14)
ylabel("Temperature $[K]$","Interpreter","latex","FontSize",14)
l = legend("\textbf{Hot gas cycle}","\textbf{Cold gas cycle}",Location="northwest
    ");
set(l, 'Interpreter', 'latex')
P01 = GE9X.amb.cruise.P0(end);
P02 = GE9X.INLET.P0_out(end);
P03 = GE9X.FAN.P0_out(end);
```

```
P04 = GE9X.COMPRESSOR.LPC.P0_out(end);
P05 = GE9X.COMPRESSOR.HPC.P0_out(end);
P06 = GE9X.CC.P0_out(end);
P07 = GE9X.TURBINE.HPT.P0_out(end);
P08 = GE9X.TURBINE.LPT.P0_out(end);
P9 = GE9X.NOZZLE.HOT.P_out(end);
P10 = GE9X.NOZZLE.COLD.P_out(end);
fig2 = figure;
plot([1 2 3 4 5 6 7 8 9],[P01 P02 P03 P04 P05 P06 P07 P08 P9],"red",[3 9],[P03 P10
    ],"blu")
xlabel("Stage","Interpreter","latex","FontSize",14)
ylabel("Pressure $[Pa]$","Interpreter","latex","FontSize",14)
l = legend("\textbf{Hot gas cycle}","\textbf{Cold gas cycle}","Location","
    northwest");
set(l, 'Interpreter', 'latex')
```